Reddit users unleashed - understanding user behaviour and their impact on meme stocks *

Simon Trimborn[†1,2] and Inez Maria Zwetsloot[3]

[1] *Amsterdam School of Economics, University of Amsterdam, Amsterdam*
[2] *Tinbergen Institute, Amsterdam*
[3] *University of Amsterdam Business School, University of Amsterdam, Amsterdam*

October 15, 2026

Abstract

Online social networks play an important role in financial markets and facilitate information sharing among investors. In the literature either theoretical models exist for the effect of individual social network users on financial markets or aggregated empirical models are tested on small networks, whereas real social networks like Reddit are large scale. We study this relationship empirically with a hand collected network dataset from WallStreetBets, a subreddit, comprising of 3.8 million posts from 502,650 users via a modelling approach we term the Interpretable Social Network and Market Behaviour modelling approach (ISMB); an approach which captures heterogeneous individual user behaviour and links this through innovative change point modelling to the stock market. We find that 1) especially followers of influential users impact market behaviour, 2) network changes occur well before the market frenzy, 3) changes in users behaviour explain the market movements and improve modelling performance.

Keywords: Social Network Market Impact, Social Network Modelling, Network dynamics, Change point detection, meme stocks

JEL classification: C55, C58, G12

*The authors are grateful for helpful discussions and comments by the seminar participants at National University of Singapore, Tilburg University, Queens Management University, University of Amsterdam, City University of Hong Kong, Stockholm Business School, SKEMA, Utrecht University and Tinbergen Institute. The audiences comments during presentation of the paper at the Spring Research Conference of ASA, Financial Econometrics meets Machine Learning Conference at Erasmus University Rotterdam, QFFE 2024 and SoFiE 2025 are gratefully acknowledged.

[†]Corresponding author

1 Introduction

The emergence of the internet enabled people all over the globe to connect with each other in ways unseen before. Through social networks, information exchange, discussions, and opinion expression became easier, more widely spread, and increasingly instantaneous. Social networks and their associated platforms also play an increasingly important role in financial markets (Deng et al., 2024; Gopal et al., 2024; Bartov et al., 2023; Peng and Zhang, 2024; Gu and Kurov, 2020). Especially Twitter received notable attention (Lehrer et al., 2021; Gu and Kurov, 2020; Behrendt and Schmidt, 2018; Yang et al., 2015), but also StockTwits (Audrino et al., 2020; Chen et al., 2022) and sentiment from across sources (Audrino et al., 2024; Gan et al., 2020; Jiao et al., 2020). However, the impact of Reddit on society and markets is less frequently studied even though the social network is the 5th most visited social media network globally as of March 2025 according to similarweb.com. In the USA, home to the largest financial market, Reddit.com is even the 2nd most visited social media network, being more popular than LinkedIn, Tiktok and X. Latest since the GameStop (GME) market frenzy in January 2021 [1] , which was driven by users of the subreddit 'Wallstreetbets' (WSB, reddit.com/r/wallstreetbets), the platform received notable attention in society and academia (Pedersen, 2022; Klein, 2022; Umar et al., 2021; Mölders et al., 2025). While attention within a community is well known to impact stock prices (Andrei and Hasler, 2014), the frenzy showed that users of social networks have real impact on financial markets. The effect of the market frenzy was so detrimental that even a US senate hearing took place, during which they questioned the main actors involved in the market frenzy, including one of the Reddit users hyping GME. In the aftermath of the GME frenzy, it became obvious that the 'internet crowd' represents a new force driving financial markets. Constantly new stocks enter the focus of the 'internet crowd', leading regularly to herding behaviour and market excesses, impacting asset prices (Aloosh et al., 2021; Costola et al., 2021). These assets range from GME, Virgin Galactic, and Palantir, to Gitlab, AMC and BB, see for example Motley Fool, Benzinga, or Yahoo Finance. Due to its continued relevance, the U.S. Securities and Exchange Commission (SEC) introduced a new term for stocks being on the radar of the internet crowd: the 'meme stock phenomenon' (SEC, 2021).

The DeGroot type model of Pedersen (2022) provides a theoretical framework for understanding (meme) stock prices discussed and traded by users of social networks. The model provides a theoretical framework for network users behaviour. It being a DeGroot model implies that users of the network change their behaviour in relation to a stock as a result of

[1] GameStop, a brick and mortar video game retailer, struggled with competition from digital distribution services and the negative economic impact of the COVID-19 pandemic. As a result, the stock price declined, and many hedge funds short-sold the stock. At that time, WSB users began to buy and hold GME stocks in bulk. Excess demand combined with limited availability drove the stock price and ultimately triggered a short squeeze. The stock price of GME rose from $ 0.70 on 3 April 2020 to $18.80 on 31 December 2020 and further to $483 on 28 January 2021. Short sellers like Melvin Capital suffered considerable losses in this short squeeze.

information they receive from the network, which translates then into price changes. However, this theoretical model is not yet thoroughly backed up by empirical evidence. It raises questions like what is driving network users' posting behaviour, which users are driving the network communication, how does this spill over into the financial markets, and how large is the effect of social network users on the actual financial market?

In this paper, we show that the change in behaviour of individual users on a social network explains market dynamics, thereby providing the empirical evidence to support the theoretical model by Pedersen (2022). For the study we hand collect 3.8 million posts from 502,650 users from WallStreetBets, a subreddit. To analyse this large scale network dataset, we propose the Interpretable Social Network and Market Behaviour modelling approach (ISMB), which combines logistic regression with network variables under sparsity and change point detection. The modelling approach captures heterogeneous individual user behaviour and links this through innovative change point modelling to the stock market. It is computationally scalable and accounts both for influential nodes as well as followers. The approach allows to study the individual relationship between any two nodes and explains their behaviour via the network variables embedded in it. This allows us to determine what drives network users' posting behaviour and to study their change in behaviour over time. We find that the change in behaviour of users on the social network aids the explanation of market movements in various ways: 1) We show that embedding the change in behaviour of users on the social network as an exogenous variable into an autoregressive model, explains a larger portion of the market dynamics than the ones without. In combination with the theory outlined in Pedersen (2022), this suggests that retail investors impact meme stock prices with their change in behaviour. 2) We find that influential users' changes in behaviour result after 6 hours in a significant effect upon market dynamics. This makes sense since a post by a single influential person first needs to reach a credible number of investors to lead too market reactions. While it is extremely challenging to detect change points in large scale networks quickly (Zwetsloot et al., 2025; Yu et al., 2022), this suggests that influential users change in behaviour could act as an early warning sign for meme stock market reactions, if identified early enough. We also find that several influential users exhibit a change in behaviour 1-2 months before the market frenzy. 3) Lesser influential users change in behaviour, if occurring in bulk, ripples instantly and with a lasting effect of various hours into the financial markets. Their change in behaviour explains the largest portion of market dynamics, even when controlling for network activity. This finding paves the way towards the construction of variables for stock market movements based on social network users behaviour.

Our proposed ISMB modelling approach was specifically designed for the modelling of individual users behaviour in large scale social networks while ensuring interpretability. Major challenges we tackle are the estimation and scalability, identification of changes in behaviour, as well as the large scale of the network with heterogeneous communications

(links). Link formation in social networks is a widely studied topic. Random network models such as Exponential Random Graph Models and Latent Variable Models have been widely studied for such tasks (Kim et al., 2018; Deng et al., 2024). Not all of these models are applicable for our purpose, as, for example, many become intractable, which makes them (only) suitable for small networks (Yon et al., 2021). Recently, there have been studies using deep learning models for link prediction (Barros et al., 2021; Yu et al., 2023; Beladev et al., 2023; Hui et al., 2024). However, these models are not easily interpretable, and therefore provide little insight into the drivers of individuals posting behaviour. For large networks another option is to use logistic regression-based dynamic network models (Agaskar and Lu, 2013; Almquist and Butts, 2013), because these provide a scalable framework for inference on large-scale temporal networks (Mallik and Almquist, 2019). In addition, these models naturally allow for incorporation of network dynamics through the autoregressive modelling structures. Furthermore, Bhattacharya et al. (2019) found that variables such as the past posting frequency and a nodes' out-degree matter for their future communication behaviour, opening the pathway to predict the probability of communication between two users based on their past behaviour and relevance to the network. This motivates our adoption of an individual-level logistic regression type model which is scalable as well as interpretable and allows naturally for incorporation of network dynamics.

The heterogeneous nature of nodes implies that their communication probabilities with other users may also be influenced by different sets of determinants. Therefore, not all parameters included in the model may be relevant for every users' behaviour, hence the system may experience sparsity. In fact, many networks are characterised by huge dimensionality, but have sparse links (de Paula, 2017). The high-dimensional nature of networks challenges the joint analysis of nodes and identification of important links. Methods from sparse modelling, originally developed for linear models (Tibshirani, 1996; Fan and Li, 2001; Zou, 2006), were increasingly used for network analysis to handle the sparse nature of the underlying models (Barigozzi and Brownlees, 2019; Trimborn et al., 2022; Trimborn et al., 2024; Zhang and Trimborn, 2023). The issue of sparsity within social networks concerns various modelling challenges, ranging from link prediction (Trimborn et al., 2024) to change point detection (Chen et al., 2024). Likewise, the ISMB modelling approach incorporates sparse estimation to account for the observed sparse structure.

Link formation in social networks is strongly interlinked with the role users have within networks. Naturally users 'trust' certain other users more, which can lead to the emergence of opinion leaders (Li and Du, 2011), also termed influencers (Han et al., 2020; Chen et al., 2022). Influencers are typically characterized by their topological position in the network (Aral and Walker, 2014), for example as measured by various centrality measures (Gelper et al., 2021). The difference in user relevance on the social network is known to explain user behaviour on a social network (Lu et al., 2013; Zhang et al., 2018) as well as its own behaviour and the one of its peers. It has also been shown that influencers do influence

one another (Ji et al., 2025). Our study builds on these findings by modelling separate models for vital and non-vital users in the network (one may think of this terminology as influencers and non-influencers), enabling us to study the effect of these different roles in the network on the markets.

We place particular emphasis on changes in the social network and propose a greedy change point algorithm, as in Bai et al. (2023), to identify changes in each link specific model. Changes are typically identified by comparing models before and after a possible change point. The overarching aim of change point detection is to identify a point in time where the network or a node in the network exhibits a change in behaviour (Ghezzi et al., 2024). Within the field of network analysis, there is an extensive literature (Kendrick et al., 2018; Woodall et al., 2017). Most of this literature considers networks with a small number of nodes, typically no more than a few thousand (Chen et al., 2024; Yu et al., 2022). The huge size of social networks in combination with their typical sparsity often challenges the existing algorithms for change point detection due to their computational complexity. In addition, most literature focuses on so-called global change point detection, identifying time points where the whole network, or a substantial part of the network, changes. An example is Zwetsloot et al. (2025) who identify change points in the same GME data, however, these are at a network community level, providing no-change identification at the nodal level. It is well known that nodes have heterogeneous effects on the network (Chen et al., 2023), especially highly influential nodes, hence the target of this study requires change point detection on the nodal level. Also, as networks are highly dynamic, users may change their behaviour multiple times within our observation period, hence we need algorithms for multiple change point detection (Niu et al., 2016; Bai et al., 2023). This motivates our adoption of multiple change points detection at the node level.

We find that the designed change point variables showed greater explanatory power for market movements than classical network measures, such as aggregated network activity. The vital users behavioural changes impact the market with a time lag of around 6 hours whereas the non-vital users impact the market immediately. Both the vital and non-vital users behavioural changes significantly relate to the integrated volatility even when controlling for aggregated network activity and established network measures proxying influential user impact. For the integrated volatility, the adjusted R^2 of a model without vs. with variables representing the behavioural change of individual users improves by 140% (from 0.182 to 0.254). We also show that the change in behaviour is not explained by market dynamics, instead it is rooted in the activity taking place on the social network.

With our study, we contribute to a number of literatures. First, we provide empirical justification for theoretical models suggesting a link between social networks and financial markets by quantifying the impact individual social network users have upon the markets (Pedersen, 2022). Second, we contribute to the literature on link prediction in large online

social networks by designing scalable node-specific link prediction models involving node behavioural and network variables (Bhattacharya et al., 2019; Lu et al., 2013; Han et al., 2020). Third, we contribute to the change point literature by proposing an algorithm for node-specific behavioural change detection and by studying the relationship between stock market movements and these changes (Chen et al., 2024; Bai et al., 2023). Fourth, we contribute to the growing literature on meme stocks. (Li and Li, 2024; Aloosh et al., 2021; Costola et al., 2021; Klein, 2022; Umar et al., 2021). Further, we provide the ISMB modelling approach which can be used for further understanding of the impact individual social network users have upon the markets beyond the effect of WSB on GME. By its design ISMB can be used not only to analyse meme stocks, but also meme coins, which emerge from similar situations like meme stocks. Hence, its applicability extends from stocks towards the cryptocurrency market.

The remainder of this paper is structured as follows. Section 2 describes the WSB network data. Section 3 introduces our model and section 4 presents the results. Finally, section 5 discusses the implications and conclusions of the study. In addition, we provide an Appendix 5 presenting computational aspects and robustness checks.

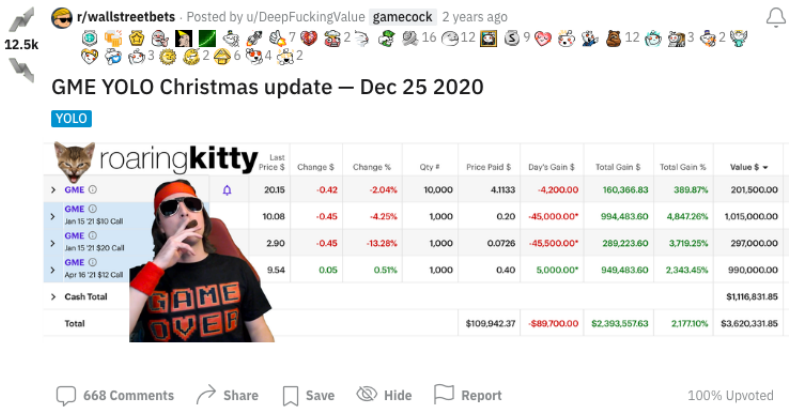
2 Empirical setting

2.1 Research context

The market frenzy of the GME stock was driven by users uniting via the subreddit Wallstreetbets, ie. /r/wallstreetbets, a subreddit focused on discussions about stocks and options trading. It started with Keith Gill, who is known on Wallstreetbets via his account 'DeepFuckingValue', posting about his long positions in GME calls. Users of WSB invested increasingly in GME via trading platforms such as Robinhood while hedge funds were betting on a falling price of GME via short positions. Many funds shorted GME while WSB investors continued to buy the stock and went long in call options, which led to a short squeeze. This situation led to a market frenzy with high daily volatility and trading volume. The David-against-Goliath hype of WSB users on Reddit led to the bankruptcy of a hedge fund, the suspension of GME trading on the popular Robinhood trading app, and an investigation by the US Congress into alleged market manipulation.

Keith Gills' posts usually show a screenshot of his portfolio and how his positions have developed, see Figure 1 for a typical post. Research covering the GME market frenzy often looked at the sentiment of users (Umar et al., 2021; Yousaf et al., 2023; Li and Li, 2024), however Figure 1 illustrates how difficult semantic analysis of the most important actor in the GME market frenzy is, given there is no text. Further, the language and culture on WSB is very different from typical finance communities, which would bias the results from

Figure 1: A typical post by Keith Gill on WSB. Gill is the most important person of the GME frenzy and is still hyped by the meme stock community. His posts usually show his portfolio and how his positions have developed. This illustrates how difficult semantic analysis of the most important actor in the market frenzy is, whereas network analysis presents itself as a viable solution.



sentiment analysis. Therefore we revert towards network analysis, given that the analysis of the existence of communication between users is robust towards the use of language, figures, gifs, and the like.

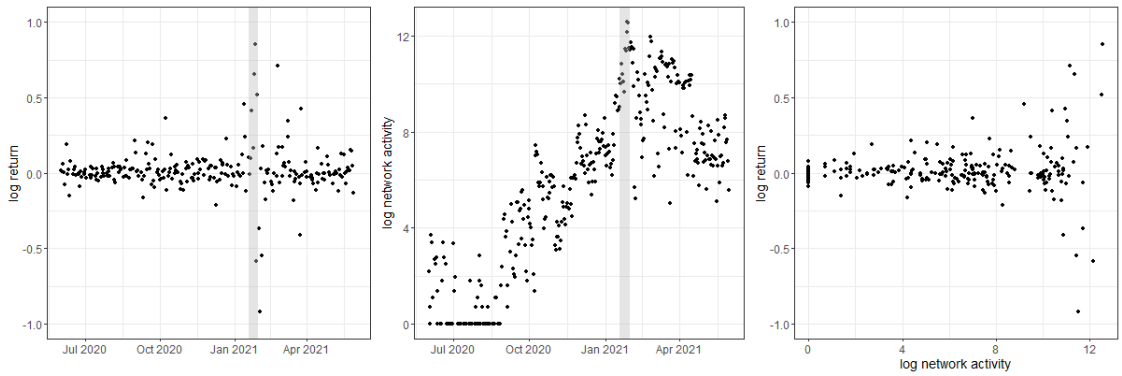
2.2 Data collection

Since the GME market frenzy, constantly new meme stocks emerge with the media picking up on them. However, during the build-up period towards the first meme stock market frenzy, the WSB network did not receive much attention and meme stocks were not a thing yet. That WSB affected the stock price of GME is well studied and GME received only little media attention during this period. This makes especially the build-up period to the GME market frenzy a suitable lab environment in which the relationship between users change in behaviour on the social network and the market dynamics of the meme stock can be studied.

To analyse the relationship between the activity on WSB and the GME stock price, we collected the discussions about GME on WSB. On Reddit, individuals submit a post on the subreddit and other members can submit comments to the post. Then others (including the original post submitter) can comment on the comments as well. Each subreddit is build around the posts, given that they are required to start a conversation.

Therefore, we first collect posts that have the keywords ‘GME’ or ‘Gamestop’ in the title. In a second step, we download the comments from the identified posts. The posting activity about GME was low before summer 2020 and the market frenzy took place in January 2021, so we collect one year of data from WSB which span the frenzy. We collected

Figure 2: Daily GME return (left Figure), daily log number of posts about GME on Wallstreetbets (middle Figure) and scatterplot between the two variables (right Figure), the frenzy period is highlighted in grey.



111,331 posts and 3,712,017 comments from 502,650 users over the time period 01.06.2020 to 31.05.2021. We visualize the relationship between the daily GME return and the daily log number of posts and comments about GME on WSB in Figure 2. We see that activity started to slowly increase in September 2020 and reached a peak at the end of January 2021, which coincides with the spikes in return in both directions for GME. In the Figure we highlight the period January 19th until January 31st 2021 as the main frenzy period. In addition to media reports, this provides a rationale for explaining GME market movements with WSB data. For more details on the webscraping and data collection process, the reader is referred to the accompanying Data-in-Brief, see Wang et al. (2024).

2.3 Network construction

The total number of users who post about GME on the WSB forum amounts to 502,650 which are our nodes in the network. If a user i commented on another user j's post (or comment) at time point t, this is modelled as a directed communication towards that user. Hence,

$$x_{ij,t} = \begin{cases} 1 & \text{if } i \text{ sends a comment to } j \text{ at time } t \\ 0 & \text{if } i \text{ does not send a comment to } j \text{ at time } t \end{cases}$$

with $x_{ij,t}$ a binary variable indicating the directed communication between i and j. If a user i submits a post or comment on their own post or comment, then this is going to be recorded as a self-communication,

$$x_{ii,t} = \begin{cases} 1 & \text{if } i \text{ submits a post or a comment to their own post/comment} \\ 0 & \text{otherwise.} \end{cases}$$

Note that self-communications are usually excluded in many network analysis studies. However, a specific aspect of the Reddit networks are the post, hence we chose to model these as self-communications allowing us to tailor the model to the specific Reddit network structure.

We observe that users commonly communicate only once per hour, hence we first split the data into hour-by-hour activity. During the market frenzy the activity is higher, hence in a second analysis, we will analyse the data 5-minute-by-5-minute during the main phase of the market frenzy.

2.4 Vital nodes

It is well established that networks have certain users who take stronger roles in the network and are central to its evolvement (Lü et al., 2016; Morone and Makse, 2015; Bamakan et al., 2019; Yang et al., 2015). In addition, communication can only happen *after* somebody has submitted a post, giving users who write posts (rather than only comments to posts) a leading role in the network. We identify vital nodes in the network in a data-driven manner. Naturally, they have to be active post submitters as a post is required to start communications in subreddits. Table 1 gives an overview of the top 10 nodes with the highest number of posts in the WSB network. Some nodes submit many posts but receive little to no response on them or did so in a very short period of time. For example, node 1075 submitted a total of 38 posts in two different weeks only and received a total of 13 comments in the posts. A high number of reactions to posts is naturally important for someone who may count as a vital node for a network. From the nodes displayed in Table 1, potential vital nodes might be 1583 and 4300, provided they are active in 14 and 30 weeks respectively and receive a huge amount of comments to their posts. The others, however, are unlikely to be influential in the network given their short window of activity and little attention received. Therefore, we set the following definition based upon activity and reactions by the network:

Definition 1. *Vital nodes are those nodes in the network that submit posts in at least 10% of the weeks and receive the highest 10% median comments on their posts.*

Notably, the definition is relaxed in its nature as it only requires posting activity in 5 weeks. The requirement for the highest 10% of median comments ensures that the users attract high attention relative to the rest of the users in the network. Using Definition 1, we identify 64 vital nodes in the network. Among them is the well known Reddit user Keith Gill with Reddit username 'DeepFuckingValue'. In terms of the overall network activity, these 64 vital nodes post only a small part of all posts (0.7%). However, nearly half of all nodes in the network have commented in at least one of their posts (45.7%) and over half of all comments is posted in a posts of a vital node (54.4%).

Table 1: Top 10 nodes with highest number of posts in the network. Potentially vital nodes might be 1583 and 4300 provided they are active in 14 and 30 weeks respectively and receive a huge amount of comments to their posts, the others however are unlikely to be influential in the network given their short window of activity and little attention received.

Node	No. posts	Median no. comments	Total no. comments	No. of weeks
3827	270	0	0	12
15725	64	0	1167	6
1583	52	30761	1811622	14
16382	47	28	4037	4
4300	45	653	137520	30
1075	38	0	13	2
46870	31	0	11	1
3194	30	0	330	6
13089	30	0	0	2
17261	29	0	1128	2

In the Appendix 5, we investigate the robustness of our Definition 1 by varying the selected percentages and making the definition more stringent. We find that the results and conclusions of our study are robust towards the selected values in Definition 1. In fact, in most instances the adj. R^2 is lowest for the settings we chose in Definition 1 (refer to the Appendix 5). Hence we denoted above definition to ensure the settings represent a conservative choice.

Note that we have selected a data driven heuristic for vital node identification rather than existing algorithms such as those reviewed by Lü et al. (2016). These algorithms are based on topological structures like closeness, centrality and degree. We did apply these type of algorithms and discovered that they do not work well for our data structure. Most do not take the dynamic structure of the network into account (properly), nor the structure of Reddit and/or are not scalable to a large network. Most algorithms identify 'vital nodes' that would only be active in a very short time period (during the frenzy) or have never submitted a post on Reddit, only comments. For a meaningful analysis, a vital node should not only pay attention to a topic when it is 'hot'. Hence, we did not see these nodes as actual influential nodes in this network and specified Definition 1. We see the development of algorithms for vital node identification in dynamic Reddit networks as an open research area.

2.5 (Admissable) Non-vital nodes

The nodes which are not vital, we will classify as non-vital. These nodes can be seen as followers of the vital nodes (influencers). It is well known that many nodes in social networks are very infrequent communicators, as is the case for the WSB dataset, see Table 2. For

Table 2: Nodes and number of ANVN in network classified by their posting behaviour. Many nodes are very infrequent communicators; only 28.1% of the nodes have posted at least 4 messages.

	No.	%
No. nodes	502,650	100%
No. nodes sending at least 1 message	498,979	99.2%
No. nodes sending at least 4 messages	141,381	28.1%
No. ANVN	48,213	9.6%
No. ANVN who follow multiple vital nodes	2778	0.6%
No. ANVN who are also a vital node	39	<0.001%

example in our WSB network, 28.1% of the nodes have posted four or more comments. Hence 71.9% of the nodes have posted less than four comments. For nodes which have so little communications, models for change in behaviour are difficult to estimate analytically. Additionally, as these nodes communicate so infrequently in the network, they do not add to its development over time.

Hence, we will focus on non-vital nodes which have sufficient communications over time and are therefore analysable in a data-driven manner. We classify nodes as *admissible non-vital node* (ANVN) of vital node i when that node has posted at least four comments in the posts of vital node i in different time periods. This is the computationally lowest possible number of communications, as otherwise the cross-validation step of our method would not be feasible computationally. Note that this defines ANVNs in relationship to specific vital nodes (VNs). It also allows for a single node to be a follower of multiple vital nodes and for posting probability estimation between vital nodes.

2.6 Network features

To understand posting behaviour, it is common to design features based on the network. Here we introduce our network based variables which we will use to predict posting behaviour (using our proposed ISMB modelling approach which will be introduced in section 3).

Macro-structure features

It has been shown that overall network activity is an important factor in modelling network influence on markets (Audrino et al., 2020; Gan et al., 2020; Jiao et al., 2020). Hence, we include the following overall network (macro-structure) feature: $AC_t = \log(\sum_{i,j} x_{ij,t} + 1)$. This measures activity at time t. Because Reddit activity is highly dependent on the hour of the day, we also include $AC_k^{24} = \log(\sum_{i,j} x_{ij,t-24*k}+1)$, which represents the log accumulated

communications posted k days prior to time point t.[2] So, when we are modelling time point t = 2PM on a Tuesday, than AC^{24}_{t-1} is the activity at 2PM on Monday and AC^{24}_{t-2} is the activity at 2PM on Sunday.

Micro-structure features

A standard feature to include in a network model, for link prediction of node i, is the in- and out-degree for node i, defined as: $D^{in}_{i,t} = \log(\sum_{j,j\neq i} x_{ji,t}+1)$, and $D^{out}_{i,t} = \log(\sum_{j,j\neq i} x_{ij,t}+1)$, where the 1 is added to avoid log(0).

As vital nodes play a key role in the Reddit network, we include a variable measuring their activity defined as $AC^{vn}_{i,t} = \log(\sum_j \sum_{h\in S_{VN}, h\neq i} x_{hj,t} + 1)$, where we exclude i from the summation to prevent possible multicollinearity issues between $D^{out}_{i,t}$ and $AC^{vn}_{i,t}$. We also include the number of comments send to the posts of the vital node i. We include this in addition to the in-degree, given communication within a vital nodes' posts but sent as a response to a comment by another user is not recorded as a message to i. This variable is defined as, $AC^{post}_{i,t} = \log(\sum_j \sum_{h\neq i} x_{jh,t} I(x_{jh,t} \in \text{post of } i))$, representing the log accumulated submits made by all users in the network at time point t to any post of node i, but not directly to i.

Hourly and market-opening-hours control variables

Social network activity is naturally dependent on the hour of the day, therefore we include a dummy variable for each hour of the day, Dum_{hour}. Also, as we are interested in analysing the network in relation to the financial markets and WSB is traded in New York, we include a dummy variable indicating if the post took place during trading hours of New York Stock Exchange (NYSE), Dum_{wls}.

3 The Interpretable Social Network and Market Behaviour Modelling Approach (ISMB)

In this section, we present the ISMB modelling approach for analysing the link between social network activity and market behaviour. The model consists of three building blocks which capture heterogeneous individual user behaviour and links this through innovative change point modelling to the stock market. In the first block, detailed in section 3.1, we build link prediction models for the individual user pairs. In the second block, detailed in

[2]We study the WSBs network on an hourly and a 5 minute level. For the 5 minute level $x_{ij,t-24*k}$ is adjusted to $x_{ij,t-24*12*k}$.

section 3.2, we are concerned with the determination of change points in the communication between users. In the third block, detailed in section 3.3, we investigate how the detected changes in user behaviour link to market movements.

3.1 Interpretable and scalable link prediction

We are interested in analysing the determinants of users posting behaviour. Therefore, we choose to model the network via link prediction models for each node pair. This individual approach enables us to estimate a dedicated model for each node pair, capturing the heterogeneous behaviour of nodes. Additionally, by design, this approach is scalable as it is easily parallelized and hence we circumvent the issue of estimating one model for a network of 502,650 nodes. We consider a parametric framework based on a logit model:

$$P[x_{ij,t} = 1] = \text{logit}(\Theta X + \epsilon_{ij,t}), \tag{1}$$

where $\text{logit}^{-1}(z) = \frac{1}{1+\exp(-z)}$ is the logistic link function applied to a linear combination of features X, extracted from the network and Θ is the parameter matrix.

To reduce the computational burden, we capitalize on the concepts of nodal importance to reduce the number of models that we will estimate. Using the classifications for vital nodes (cf. section 2.4) and admissible non-vital nodes (cf. section 2.5), we define a set of vital nodes S_{VN} and a set of ANVNs, S_{ANVN}. Then we estimate our model (Eq. (1)) for $j \in S_{VN}$ and $i \in S_{VN} \cup S_{ANVN}$. This results in a total of $51,245$ models to estimate, which is a tremendous reduction from the $n^2 = 502,650^2$ possible models to fit.

To estimate equation (1) we need to select features to include in the X matrix. An overview of the included features is given in Table 3, which were detailed in section 2.6. This includes variables representing the VNs' and ANVNs' behaviour, such as posting frequency (out-degree), in-degree, and the nodes' past behaviour, in line with findings of Bhattacharya et al. (2019), Lu et al. (2013), and Yu et al. (2022). Further, we include overall network activity, which is known to be an important factor in modelling social network influence on markets. Also, behaviour of people is observed over time and their reaction to posts may have a time lag, therefor we include K lags of all features. In addition, we include a set of control variables.

Table 3: Features included in X. We include two groups of network dependent features: micro-structure, macro-structure as well as control variables. All features, except the controls, are included with K lags as part of our link prediction model (equation 1) for VNs and ANVNs.

Type	Variable	Description	VN and/or ANVN model
Dependent	$x_{ij,t}$	Comment send from i to j	both
Micro-structure	$D^{in}_{i,t}$	In-degree of i: number of comments send to i	both
	$D^{out}_{i,t}$	Out-degree of i: number of comments send by i	both
	$D^{in}_{j,t}$	In-degree of j: number of comments send to j	ANVN
	$D^{out}_{j,t}$	Out-degree of j: number of submits by j	ANVN
	$AC^{vn}_{j,t}$	No. of submits by all other vital nodes (excluding j)	both
	$AC^{post}_{j,t}$	No. of comments within any post of VN j	both
Macro-structure	AC_t	Overall network activity	both
	AC^{24}_k	Overall network activity k days prior to time t	both
Control	Dum_{wls}	Dummy indicating trading hours	both
	Dum_{hour}	24 dummies for each hour	both

Hence, we set up the following 64 models, one for each VN:

$$\begin{aligned}\text{logit}(AX+\epsilon_{ii,t}) = \text{logit}(\theta_0 + \sum_{k=1}^{K}[&\theta_{1,k}x_{ii,t-k} + \theta_{2,k}D^{in}_{i,t-k} + \theta_{3,k}D^{out}_{i,t-k} \\ &+\theta_{4,k}AC^{vn}_{i,t-k} + \theta_{5,k}AC^{post}_{i,t-k} + \theta_{6,k}AC_{t-k} + \theta_{7,k}AC^{24}_k] \\ &+\theta_8 Dum_{wls} + \sum_{h=1}^{24}\theta_{9,h}Dum_{hour,h} + \epsilon_{ii,t}),\end{aligned} \tag{2}$$

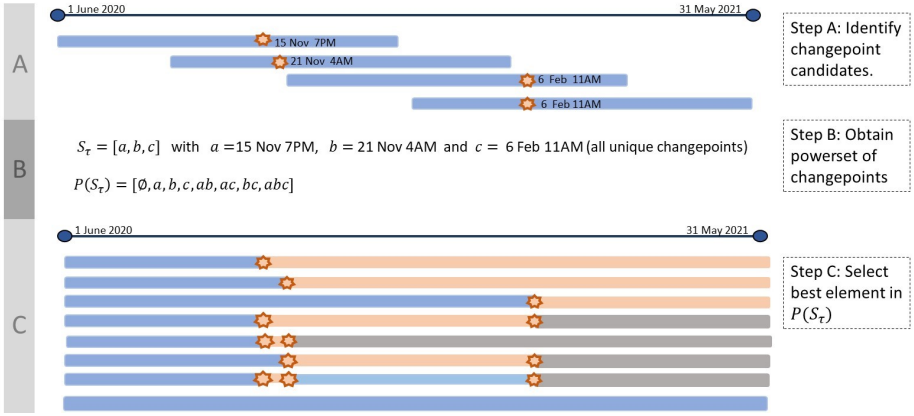
where K is the number of lags included in the model. For the admissible non-vital nodes and their communication with a vital node, we set up the following 51,181 models, one for each relevant ANVN-VN pair:

$$\begin{aligned}\text{logit}(AX+\epsilon_{ij,t}) = \text{logit}(\theta_0 + \sum_{k=1}^{K}[&\theta_{1,k}x_{ij,t-k} + \theta_{2,k}D^{in}_{i,t-k} + \theta_{3,k}D^{out}_{i,t-k} + \theta_{4,k}D^{in}_{j,t-k} \\ &+ \theta_{5,k}D^{out}_{j,t-k} + \theta_{6,k}AC^{vn}_{j,t-k} + \theta_{7,k}AC^{post}_{j,t-k} + \theta_{8,k}AC_{t-k} \\ &+ \theta_{9,k}AC^{24}_k] + \theta_{10}Dum_{wls} + \sum_{h=1}^{24}\theta_{11,h}Dum_{hour,h} + \epsilon_{ij,t}).\end{aligned} \tag{3}$$

3.2 Changes in behaviour via change point identification

It is well known that social networks are highly dynamic and that individuals change their posting behaviour over time (Aral and Walker, 2014; Bhattacharya et al., 2019; Deng et al., 2024). In addition, Pedersen (2022) motivates that the change in network behaviour can

Figure 3: The three step algorithm for change point identification. In Step A, a rolling window approach is being used to identify one change point per window. In Step B, the powerset of these change points is being constructed to identify the potential change point combinations. In Step C, the best combination of change points as by maximum likelihood of the underlying models is being identified while including the possibility of having no change point.



spill over into financial markets. To link individual change behaviour to the stock market behaviour, we have chosen to conduct a change point approach to identify if and when users change their behaviour. For the model in equation (1), we consider a change point structure:

$$P[x_{ij,t} = 1] = \sum_{s=1}^{S+1} \text{logit}(\Theta_s X + \epsilon_{ij,t}) \mathbb{1}(\tau_{s-1} < t \leq \tau_s), \tag{4}$$

where S equals the number of change points that occur at time points τ_s, where $1 < \tau_1 < \tau_2 < ... < \tau_S \leq T$ and $\mathbb{1}(\cdot)$ is the indicator function. We set $\tau_0 = 1$ and $\tau_{S+1} = T$.

We specify a change point detection framework. The method is based upon a greedy algorithm for searching for multiple change points in time series data. The method works by first identifying candidate change points by searching for a single change point in a rolling window and then combine these into possible change point sequences, as done in e.g. Bai et al. (2023). Our implementation of this general framework is visualized in Figure 3. The algorithm we propose consists of three steps (A-C), is entirely data driven, and is implemented as an exhaustive search algorithm:

Step A: Identify change point candidates by conducting a rolling window search and identifying a single (possible) change point in each window.

Step B: Obtain all possible change point sequences by constructing a set of unique change points and building the powerset.

Step C: Select the best change point sequence by estimating a model for each combination of change points and selecting the best model and related change point sequence.

We provide a detailed description for steps A-C in the Appendix 5. The resulting model with the identified change points typically has hundreds of parameters, challenging the interpretability of the parameters. To tackle this issue, we perform LASSO regularization (Tibshirani, 1996), which is an often applied technique in network analysis (Barigozzi and Brownlees, 2019; Trimborn et al., 2022; Trimborn et al., 2024; Zhang and Trimborn, 2023). The LASSO enables us to estimate sparser models, allowing for an interpretable model.

To evaluate whether our model with change points (i.e. equation (4)) statistically significantly outperforms a model without change points (i.e. equation (1)) we compute the Vuong test (Vuong, 1989) for overlapping models, as the models potentially overlap. We only include models with change points if they show a better fit at a 5% significance level.

3.3 Linking the network to the market

To determine the validity of the hypothesis that social network users change in behaviour affects the markets, we analyse if the change in individual users' posting behaviour on the social networks relates to stock market activity. For this we first define the variables CP_t^{VN} and CP_t^{ANVN}, measuring the number of change points occurring at time point t of the VNs and ANVNs:

$$CP_t^{VN} = \log\Big(\sum_{j\in S_{VN}, i=j}\sum_{r=1}^{R}\mathbb{1}(cp_r^{ij}=t)\Big) \tag{5}$$

$$CP_t^{ANVN} = \log\Big(\sum_{j\in S_{VN}, i\in S_{VN}\cup S_{ANVN}, i\neq j}\sum_{r=1}^{R}\mathbb{1}(cp_r^{ij}=t)\Big). \tag{6}$$

where cp_r^{ij} with $r = 1, .., R$ are the selected change points for node pair i and j (ie. the output from step C).

We study the market activity by considering returns, volatility, trading volume and option trading volume. As the changes in users behaviour are observed at an hourly frequency, we consider hourly returns. From 5-minute price data for GME, we construct the realized volatility for every hour. Provided GME experienced large price swings with intraday jumps, we split the volatility into the integrated volatility (IV) and the jump component (JV) with the Andersen et al. (2007) estimation procedure. Trading Volume is observed as an aggregate over the past hour. Since a lot of trading activity from WSB users occurred on option markets, we also analyse the Option Trading Volume aggregated over the past hour. Due to the high market activity surrounding the frenzy, we take the measured 5-minute frequency changes in behaviour of the WSB users and also analyse this effect during the frenzy. For this, we derive returns, integrated volatility, and jump volatility for 5-minute frequency from 1-minute return observations as well as observe trading volume and option trading volume at 5-minute frequency.

We estimate the following model:

$$M_t = \alpha_0 + \sum_{k=1}^{K} \alpha_k M_{t-k} + \sum_{k=0}^{K} \beta_k^{VN} CP_{t-k}^{VN} + \sum_{k=0}^{K} \beta_k^{ANVN} CP_{t-k}^{ANVN} + \sum_{k=0}^{K} \beta_k^{AC} AC_{t-k} + \sum_{k=0}^{K} \sum_{i=1}^{M} \beta_k^i w_{t-k}^i + \epsilon_t. \tag{7}$$

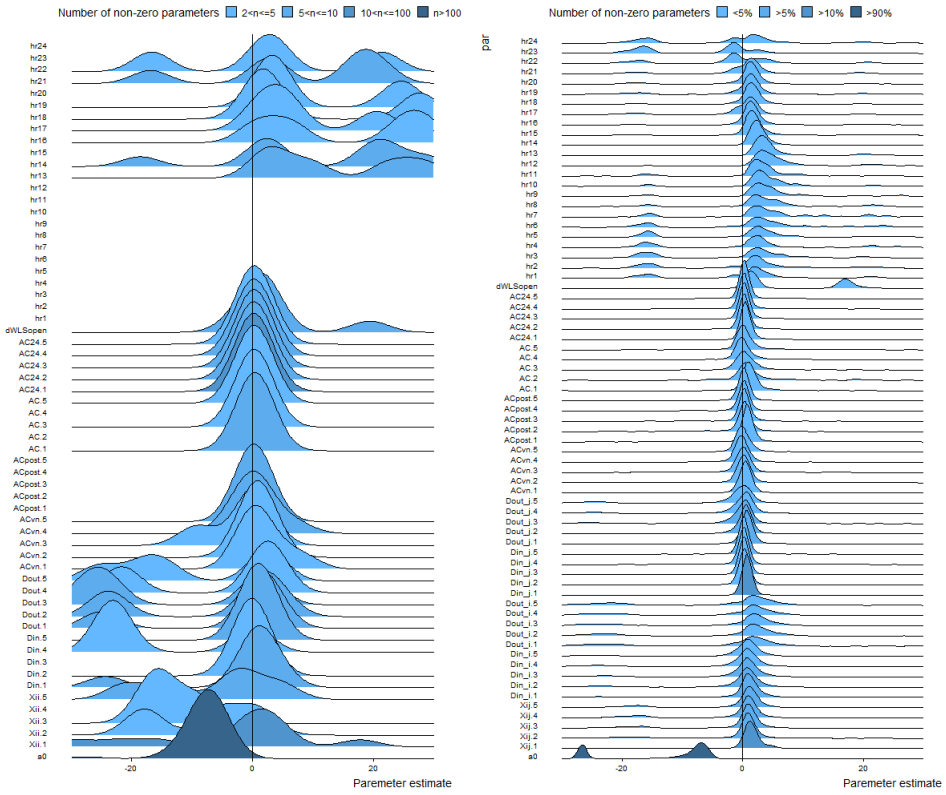
where M_t is the market variable at time t, which is either the return (r_t), integrated volatility (IV_t), jump volatility (JV_t), trading volume (TV_t), or the option trading volume (OTV_t). The variable CP are the change point variables on a log scale, as defined in equations (5) and (6), AC is the overall network activity on a log scale, a well established predictor for market movement. Further W is a $T \times p$ matrix consisting of p network related variables, which are included as control variables. As variables in W, we consider various variables which measure the network activity and importance of nodes. Specifically, we include the log mean distance, log mean degree and the log number of active nodes. Note that jumps are assumed to appear independent of each other, by the nature of their definition, hence for the jump volatility model we exclude time-lagged JV variables from equation 7. This allows us to investigate if the addition of contemporaneous and past changes in behaviour of vital and admissible non-vital nodes has an impact on the market.

4 Results

To understand the microstructure in the WSB network, we analyse the network data, described in section 2, with our proposed ISMB modelling approach described in section 3. We run ISMB with a window size of 2 months ($h = 1440$ hours) and the window rolls forward with a step size of 1 week ($l = 168$ hours). To facilitate the model estimation, we set the minimum time between change points to 1 week ($k = 168$). Recall that our ISMB modelling approach consists of three blocks. In section 4.1, we report on the modelling results of the first block focused on the determinants of users communication. In section 4.2, we report on the results of the second block focused on changes in behaviour. In the analysis we only consider the change points which are significant based on the Vuong test. We also report on deviance and accuracy. These results highlight the importance of ANVNs in the network evolution. In section 4.3, we report on the results of the third block focussed on linking the network to the market. Here we show that the individual users behaviour on Wallstreetbets influenced the market activity of the GME stock.

In addition, to get a detailed view of the network around the market frenzy, we also study the network at a 5-minute-by-5-minute frequency for the period of one month around the frenzy in section 4.3.2. In section 4.4 we investigate for the possibility of reverse causality, meaning if the market dynamics lead to the change points instead of vice versa. Finally, in section 4.5, we comment on the economic and financial implications of our findings.

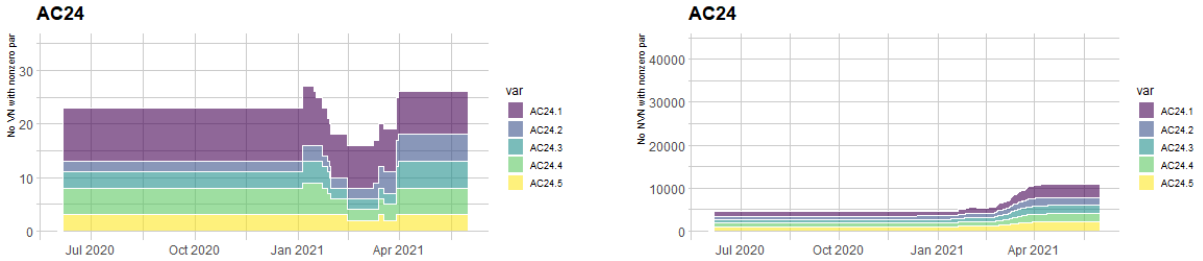
Figure 4: Value of the non-zero parameters for the VNs (left) and ANVNs (right). Darker kernels indicate more models with non-zero parameters. During trading hours (hr 14 until hr 20, UTC) there is an increases probability of posting. After comments an ANVN is likely to comment again indicated by positive D_{out} and x_{ij} parameters.



4.1 The determinants of users communication

Next, we are going to analyse the parameters of the models. As discussed in section 3, we use LASSO to ensure sparsity and interpretability in the models we estimate. For the VN and ANVN models, we show the value of the non-zero parameters in Figure 4. We observe that during trading hours the VNs have an increased probability of posting (hr 14 until hr 20). Note, we define time in UTC time zone. For the ANVNs, the effect of hours is more dispersed, which is expected as there are many more individuals. Next, we look at the dummy for trading hours (Dum_{wls}). Both VNs and ANVNs have predominantly positive values for this parameter, indicating that posting is more likely during market trading hours. There is a group of VNs and ANVNs that have high values (around 20) for the trading hour dummy, indicating a strong likelihood of posting during trading hours. For the VNs, the overall activity in the network seems to have no specific directional effect on the likelihood of posting, since parameter values are centered around zero. On the contrary, for the ANVNs the overall network activity has parameter distributions skewed to the right of the zero line, indicating a slight increase in posting likelihood of nodes, when the overall

Figure 5: Left: VNs AC_{24}, Right: ANVNs AC_{24}



network is active. As expected, D_{out} for the VNs and $D_{out,i}$ for the ANVNs is positive for the majority of the estimated parameters, indicating that when users post, they are likely to post again. However, there is also a group of users for which this parameter has a negative value around -20 to -25, indicating that these people tend to post and then probably leave the platform to come back at a later time. Finally, we consider for the VNs the effect of X_{ii}, which is mostly negative, indicating that after submitting a post, the VNs are less likely to submit another post. When we consider the ANVNs and the parameters of the variable X_{ij}, we see the opposite effect. When ANVNs submit comments to posts, they are more likely to comment again in the post.

We observe in Figure 4 that AC^{24} mattered for a lot of VNs, so we compare it over time and against its relevance for the ANVNs. Figure 5 displays for how many vital nodes the respective parameter plotted was different from 0 over time. In Figure 6, we show various parameters which are of high importance for the ANVNs. We observe that for the VNs the activity of the network is an important variable determining their posting probability. The lagged activity AC^{24} is relevant for up to 39 vital nodes. Its relevance decreases during the market frenzy but sharply increases right after. Even though ANVNs are slightly positively impacted by network activity, only a fraction of them is impacted by it, see Figure 6. It only increases after the market frenzy. As we see from Figure 4, VNs are further impacted by their own posting behaviour and posts of others reaching them. Some of them post again once they have posted or were contacted, but also various do not post in streaks. The ANVNs show more interesting behaviour, as the bulk of parameters associated with communication are positive, compare Figure 4.

Communication initiated by the ANVNs, compare Figure 6, mattered but the communication pattern of the VN they are connected with, mattered a lot more. If the VN communicated, for 30% of ANVNs, this mattered for their own posting probability. Likewise, if the VN received messages, it affected the ANVNs communication probability instantly, see Figure 6. About 33% of ANVNs were affected when their VN received a message. With the onset of the market frenzy, the VNs became even more important for them, with 45% of ANVNs being affected by their communication pattern. However, the most important

Figure 6: ANVNs - Upper Left: Din_i, Center: Din_i, Right: X_{ij}, Lower Left: $Dout_i$, Right: $Dout_j$

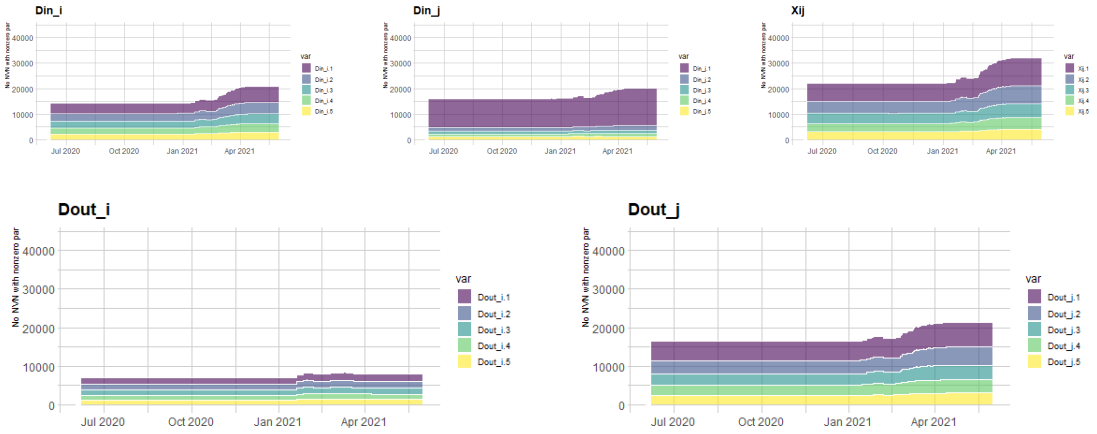


Table 4: Number of identified change points after applying our proposed ISMB modelling approach to the WSB network. We find that the majority of the VNs and ANVNs change their posting behaviour once.

No. cps	VN	%	ANVN	%
0	13	20.3	14339	28.02
1	50	78.1	36618	71.55
2	1	1.6	210	0.41
3	0	0.0	11	0.02
≥ 4	0	0.0	0	0.00

factor was direct communication between the VN and ANVN. Before the market frenzy, it affected the communication probability of 46% of ANVNs and increased with the market frenzy until it affected 64% of ANVNs. This indicates a strong change in many ANVNs behaviour towards being a lot more affected by what the VNs they follow do, and also towards being impacted by posts about GME from the general network. This shows that VNs are a driving force in the network.

4.2 The importance of admissible non-vital nodes

Next we study the change point results of ISMB. These change points identify the changes in users believes which in the theory of Pedersen (2022) is the link between the social network and the markets. First, we analyse the identified change points in our models, reported in Table 4. Note that these change points are statistically significant at a 5% level as tested with the Vuong test. We see that most VNs and ANVNs change their posting behaviour once in the year (78% and 72%) whereas 20% of the VN and 28% of the ANVN do not have any change points in their models. The maximum number of change points we find is

Figure 7: Identified significant change points over time for VN (left) and ANVN (right). Some VNs change their behaviour as early as November 2020 which is well before the frenzy. On the contrary, most ANVNs change their behaviour during and after the frenzy.

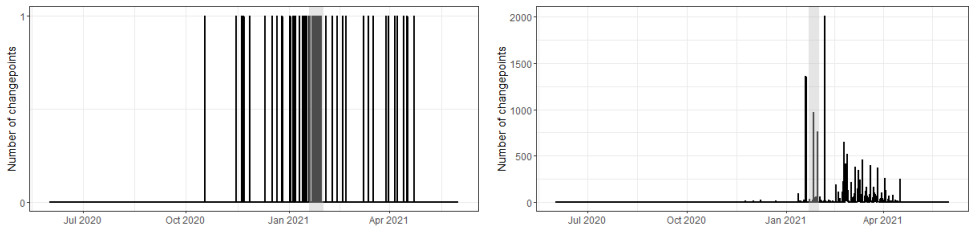
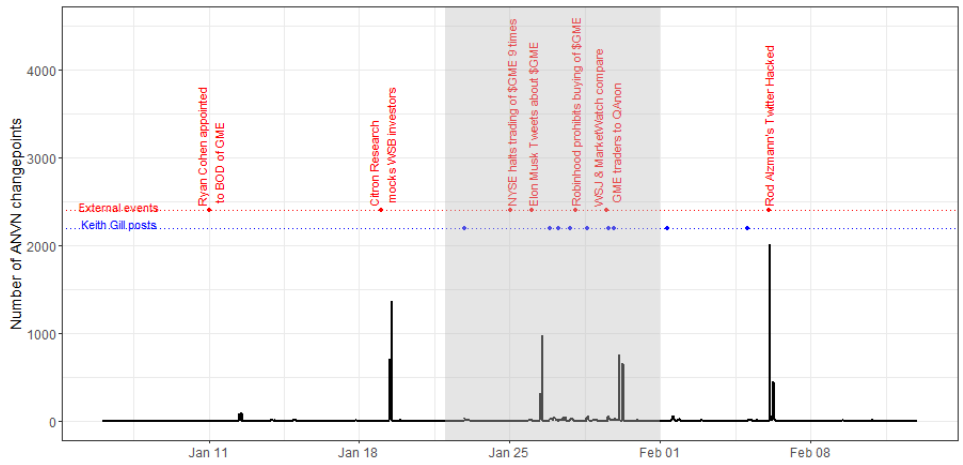
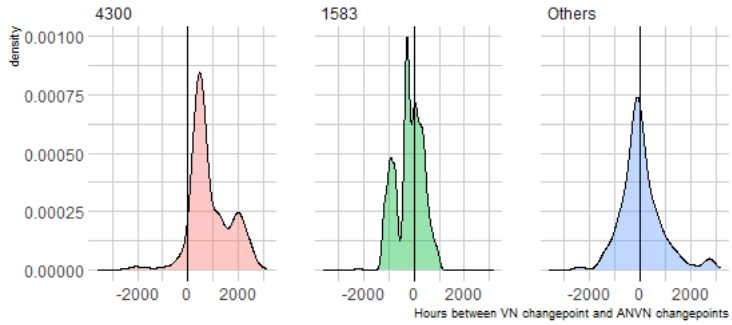


Figure 8: Change points over time for ANVNs during frenzy period labelled with external events



3. Figure 7 shows the change points of all VN (left) and all ANVN (right) over time. We see that some VNs change their behaviour as early as November and December 2020 which is well before the frenzy. On the contrary, most ANVNs change their behaviour during and after the frenzy. In Figure 8, we zoom in on the period around the frenzy and look at the ANVNs' change points. There are large peaks at specific moments in time, which indicate hours during which many ANVNs (up to 2000) show change in their posting behaviour. Figure 8 also indicates key events (taken from *GameStop Stock Price Timeline* 2024) and the posts of Keith Gill. We see that various events, but not all, coincide in time with large peaks in the number of change points we find for the ANVN models. Whereas studies such as Gan et al. (2020) highlight the relevance of overall network activity and Yang et al. (2015) show influential nodes like the VN are most important, this highlights the importance of monitoring individual and less important users change in behaviour. It also shows that monitoring news instead of the network would have missed or overestimated the relevance of events, given users reacted strongly to some and others only slightly. Occurring in bulk and even when controlling for network activity, they lead to market reactions as we show in section 4.3.

Figure 9: Time between change points of VNs and change points of ANVNs.



To study in more detail the reaction of ANVNs to VNs change points, we look at the time between the VNs (first) change point and the ANVNs (first) change point. Figure 9 shows a density plot of these times for the most active vital nodes 4300 and 1583 as well as all the other VNs. We see that especially for VN 4300, the change in posting behaviour seems to result in ANVNs changing their posting behaviour a while later, with the peak of the distribution at 400 hours (16 days) after VN 4300 changed its behaviour. For VN 1583, a well-known moderator on WSB, the effect is less pronounced and inline with the role of moderators (He et al., 2024) which are more reactive to the network. Again, this indicates that only looking at important nodes such as VNs does not suffice.

We evaluate model deviance and accuracy to understand how well our model fits and predicts the users posting behaviour and how important it is to include change points in our modelling approach. Figure 10 shows the percentage decrease in deviance where we compare our model to a model with intercept-only and without change points. By this, we investigate if the variables and change point detection method we suggest, improves the prediction of users communication behaviour. We see that our model decreases the deviance substantially. Figure 11 shows the percentage points increase in prediction accuracy of our model compared to the intercept model without change points. We only consider the predictions of communications (ie. the 1's in our data) as the data is highly unbalanced. For the VNs, we see a small group of models for which we predict all communications accurately and overall the accuracy is better for the models with change points than the base line model without change points. This shows the outperformance of our modelling framework for modelling and analysing individual users posting behaviour.

4.3 Impact of Wallstreetbets on GME stock price

In this subsection, we link the user behavior to the financial markets and study the affect on the return and volatility of the GME stock. We first perform a full sample analysis on

Figure 10: Relative decrease in deviance for our model compared to an intercept model without change points. Left: VNs, Right: ANVNs

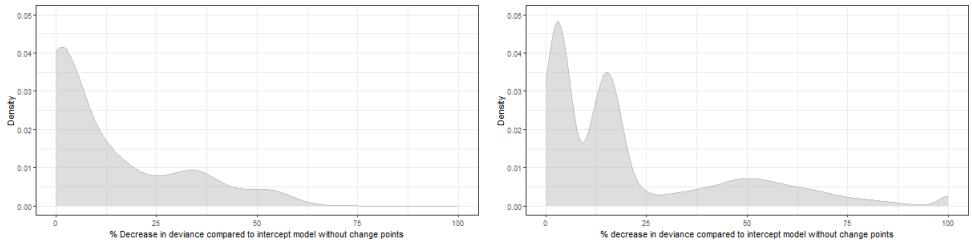
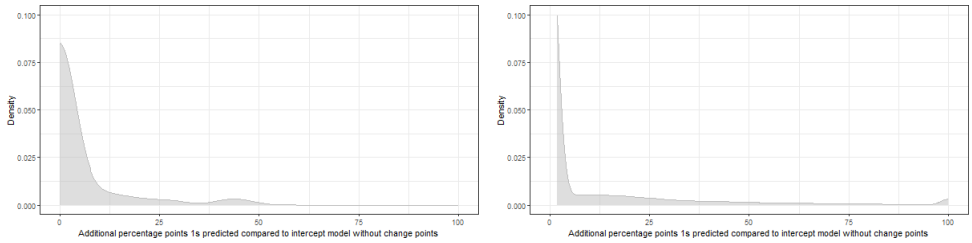


Figure 11: Percent points increase in prediction accuracy for our model compared to an intercept model without change points. Left: VNs, Right: ANVNs



an hour-by-hour basis followed by a detailed study of the frenzy period on a 5minute-by-5minute basis.

4.3.1 Hourly analysis: Changes in users behaviour and market impact

We conduct a full sample analysis, which allows us to analyse the parameters driving the stock dynamic. We observe for the VNs and ANVNs a significant effect on the return, integrated variance (IV) and jump volatility (JV) of GME, compare Tables 5, 6, and 7. We observe only little effect onto trading volume (TV), see Table XXX. It is well documented that users of WSB primarily traded options, for which we observe a stronger predictability, see Table XXX. However, the improvement is not as strong as for return, IV, and JV. When adding the VN change points, a significant parameter occurs in lag 6 for IV and JV. This infers that it takes 6 hours until the change in behaviour of VNs is reflected in the integrated variance and jump volatility of the GME stock. This is reasonable given the change in behaviour will be accompanied by a post on WSB and it would take a while until sufficient users have seen the post, reacted to it and (potentially) acted based on it on the stock market. For the return, we observe a faster effect from the VNs, materialising already after 3 hours. As for the $ANVNs$, we see an even more drastic increase in the adjusted R^2, and various variables over several lags are significant. Notably, the market reacts faster to their changes in behaviour. We observe this from earlier lags being significant. Given the different roles of VNs and ANVNs in the network, it is reasonable that the effect of ANVNs is reflected faster in the market. Whereas the effects of the change points of VNs and ANVNs

Table 5: Modeling returns with equation (7). Columns show different combinations of included features. For the sake of brevity insignificant variables and significant lagged return variables are not displayed.

	Dependent variable: return					
	(1)	(2)	(3)	(4)	(5)	(6)
VN3			-0.021^{***}	-0.020^{***}		-0.022^{***}
			(0.007)	(0.006)		(0.006)
VN4			0.006	0.012^{**}		0.008
			(0.006)	(0.006)		(0.006)
NVN				-0.004^{***}		-0.003^{***}
				(0.001)		(0.001)
NVN1				0.007^{***}		0.008^{***}
				(0.001)		(0.001)
NVN2				0.002^{**}		0.003^{***}
				(0.001)		(0.001)
NVN3				-0.004^{***}		-0.004^{***}
				(0.001)		(0.001)
NVN6				-0.001^{*}		-0.001^{*}
				(0.001)		(0.001)
Constant	0.0003	0.0002	0.0002	0.001	0.0003	0.001
	(0.0004)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Lagged returns	Yes	Yes	Yes	Yes	Yes	Yes
Vital nodes	No	No	Yes	Yes	No	Yes
Non-vital nodes	No	No	No	Yes	No	Yes
Network activity	No	Yes	Yes	Yes	Yes	Yes
Network statistics	No	No	No	No	Yes	Yes
Observations	1,633	1,633	1,633	1,633	1,633	1,633
R^2	0.009	0.020	0.028	0.081	0.045	0.124
Adjusted R^2	0.004	0.011	0.014	0.064	0.021	0.093
Note:						*p<0.1; **p<0.05; ***p<0.01

on predictability is pronounced for return, IV, and JV, the same is not true for trading volume and option trading volume. The users of WSB are known for trading options instead of on spot markets, hence it is unsurprising that the effect upon trading volume is minor, see Table XXX. The predictability of option trading volume is generally higher, see Table XXX, but including change points of VNs and ANVNs does not improve the adj. R^2. These results show that the volume users traded was not affected by users change in behaviour, suggesting that the change in behaviour did not bring in additional trading activity, while affecting returns and volatility (IV, JV) though. All models containing VNs and ANVNs change points as variables, also contain the network activity as a variable. In light of this, it is worth pointing out that the network activity AC does not proxy the individual users change in behaviour, but remains statistically significant for the market predictability. This shows that the change points measure user behaviour the overall network activity cannot represent. Further, it is not only the majority over users who matter, but also the influential users, since the change in behaviour from VNs remains important for the explainability of the return, integrated variance, jump volatility, and trading volume.

Table 6: Modeling integrated variance (IV) with equation (7). Columns show different combinations of included features. For the sake of brevity insignificant variables and significant lagged IV variables are not displayed.

	Dependent variable: integrated variance (IV)					
	(1)	(2)	(3)	(4)	(5)	(6)
VN6			0.064^{***}	0.065^{***}		0.061^{***}
			(0.006)	(0.006)		(0.006)
NVN				-0.003^{***}		-0.004^{***}
				(0.001)		(0.001)
NVN1				0.005^{***}		0.004^{***}
				(0.001)		(0.001)
NVN2				0.001		0.002^{**}
				(0.001)		(0.001)
NVN3				0.002^{*}		0.003^{***}
				(0.001)		(0.001)
NVN4				-0.002^{**}		-0.002^{*}
				(0.001)		(0.001)
NVN5				-0.003^{***}		-0.003^{***}
				(0.001)		(0.001)
Constant	0.001^{***}	-0.002^{**}	-0.001^{**}	-0.001^{*}	-0.0004	-0.0004
	(0.0004)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Lagged IV	Yes	Yes	Yes	Yes	Yes	Yes
Vital nodes	No	No	Yes	Yes	No	Yes
Non-vital nodes	No	No	No	Yes	No	Yes
Network activity	No	Yes	Yes	Yes	Yes	Yes
Network statistics	No	No	No	No	Yes	Yes
Observations	1,633	1,633	1,633	1,633	1,633	1,633
R^2	0.175	0.189	0.239	0.265	0.255	0.328
Adjusted R^2	0.172	0.182	0.228	0.250	0.237	0.304
Note:						*p<0.1; **p<0.05; ***p<0.01

The explainability of the returns is remarkably high when considering the users change points in the model. Commonly asset returns experience a lot lower R^2. To explore the dynamics over time further, we conduct a rolling window analysis. We apply model (7)

Table 7: Modeling integrated variance (IV) with equation (7). Columns show different combinations of included features. For the sake of brevity insignificant variables are not displayed.

	Dependent variable: Jump volatility (JV)				
	(1)	(2)	(3)	(4)	(5)
VN6		-0.028^{***}	-0.028^{***}		-0.028^{***}
		(0.007)	(0.008)		(0.007)
NVN			-0.003^{**}		-0.003^{**}
			(0.001)		(0.001)
NVN1			0.006^{***}		0.006^{***}
			(0.001)		(0.001)
NVN3			-0.002^{**}		-0.002
			(0.001)		(0.001)
NVN5			0.002^{**}		0.002^{**}
			(0.001)		(0.001)
Constant	-0.001	-0.001^{*}	-0.001	-0.002^{**}	-0.002^{*}
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Vital nodes	No	Yes	Yes	No	Yes
Non-vital nodes	No	No	Yes	No	Yes
Network activity	Yes	Yes	Yes	Yes	Yes
Network statistics	No	No	No	Yes	Yes
Observations	1,633	1,633	1,633	1,633	1,633
R^2	0.022	0.032	0.045	0.102	0.130
Adjusted R^2	0.017	0.022	0.031	0.084	0.103
Note:					*p<0.1; **p<0.05; ***p<0.01

Table 8: Significant variables from models 13 with differing variables include and Lag length 7. Insignificant variables are not displayed. Control variables (network statistics and network activity) are included for differing models.

	Dependent variable: Trading Volume (TV)					
	(1)	(2)	(3)	(4)	(5)	(6)
VN2			1.762^{***}	1.798^{***}		1.784^{***}
			(0.350)	(0.352)		(0.355)
Constant	0.046^{**}	0.057	0.057^{*}	0.051	0.057	0.058
	(0.020)	(0.035)	(0.035)	(0.037)	(0.042)	(0.042)
Lagged TV	Yes	Yes	Yes	Yes	Yes	Yes
Vital nodes	No	No	Yes	Yes	No	Yes
Non-vital nodes	No	No	No	Yes	No	Yes
Network activity	No	Yes	Yes	Yes	Yes	Yes
Network statistics	No	No	No	No	Yes	Yes
Observations	1,633	1,633	1,633	1,633	1,633	1,633
R^2	0.012	0.014	0.031	0.034	0.026	0.047
Adjusted R^2	0.007	0.005	0.017	0.016	0.002	0.014
Note:						*p<0.1; **p<0.05; ***p<0.01

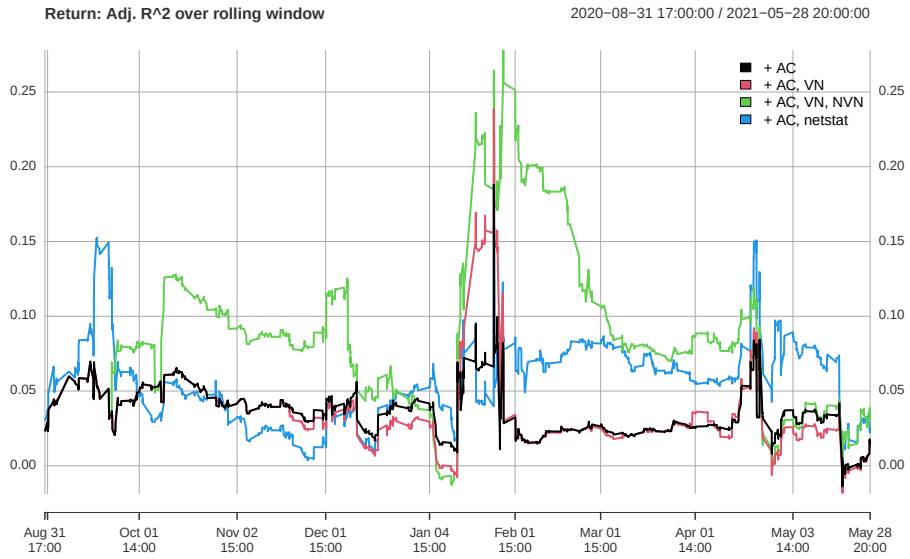
Table 9: Significant variables from models 13 with differing variables include and Lag length 7. Insignificant variables are not displayed. Control variables (network statistics and network activity) are included for differing models.

	Dependent variable: Option Trading Volume (OTV)					
	(1)	(2)	(3)	(4)	(5)	(6)
NVN				-0.061^{*}		-0.070^{*}
				(0.036)		(0.036)
Constant	-0.001	-0.054	-0.055	-0.072^{**}	-0.079^{**}	-0.095^{**}
	(0.020)	(0.034)	(0.034)	(0.036)	(0.039)	(0.040)
Lagged Option TV	Yes	Yes	Yes	Yes	Yes	Yes
Vital nodes	No	No	Yes	Yes	No	Yes
Non-vital nodes	No	No	No	Yes	No	Yes
Network activity	No	Yes	Yes	Yes	Yes	Yes
Network statistics	No	No	No	No	Yes	Yes
Observations	1,687	1,687	1,687	1,687	1,687	1,687
R^2	0.209	0.215	0.217	0.221	0.233	0.240
Adjusted R^2	0.205	0.208	0.206	0.206	0.215	0.214
Note:						*p<0.1; **p<0.05; ***p<0.01

during the rolling window analysis on 3 months of data and roll forward with step width of 1 hour. Figure 12 summarises the adjusted R^2 on the rolling windows of 4 different model specifications, each containing a different subset of the variables of interest. The black line contains the benchmark model consisting of the time series own past (7 lags of returns) and aggregated network activity (AC), contemporaneous and also with 7 lags, as explainable variables. This number of lags were chosen as they span the past trading day, i.e. if we model the return at 11 am, 11.01.21, then 7 lags span back until 11 am, 10.01.21, since we only consider market opening hours. The red line adds the VN change points to the model and the green one the ANVN change points. To compare if the change points of the VN and ANVNs contain additional information, which cannot be captured by regular network statistics (log mean distance, log mean degree, log number of active nodes), the blue line shows the adjusted R^2 of the model containing AC and the network statistics.

We observe from Figure 12, that the change points of the VN and ANVN add to the explainability of the GME return dynamics. Before the market frenzy, from August to December, the model containing AC, VN and ANVN change points (green), results in much higher adjusted R^2, up to 0.12. For return predictability, this is already remarkably high. This suggests that the VN and ANVNs were already a decisive force for the price dynamics. Note that the model containing the returns own past and AC, as well as the network statistics additionally, result in much lower adjusted R^2, between $0 - 0.05$. Interestingly, the adjusted R^2 drops strongly for all models in December, during the building up period of the frenzy. This may suggest that during this brief period, the former notable explainability from VN and ANVNs change in behaviour was priced into GMEs stock price. During the market frenzy itself, the return explainability reaches remarkably high adjusted R^2 of up

Figure 12: Adj. R^2 for return models with rolling window over 3 months with step width 1 hour. Change points of the VN and ANVN add to the explainability of the GME return dynamics.



to 0.28. The return predictability from the users' changes in behaviour remains very high during the entirety of the market frenzy, showing the impact the social network had on the market during that time. It is further remarkable, that classical network measures such as AC and the network statistics were not able to explain the market movements that well. The explainability of the return dynamics from the changes of users behaviour remained high also after the frenzy, outperforming all other models, showing that the social network had a longer lasting effect on the markets, going beyond the main market frenzy.

In summary, these results show that the changes in users behaviour have a strong and long lasting effect on the return dynamics. In particular, changes in ANVN-users behaviour excel in doing so, VN-users play a lesser role in explaining the return dynamics. This implies that not only highly influential users, despite inciting communication in the network, had an effect on the market. Their impact is comparably less pronounced than the changes in behaviour emerging from the masses of users on the social network.

4.3.2 5-minute analysis: The network during the market frenzy

A particular period of interest for GME is the time spanning the market frenzy. During this time we also observed that a lot more activity took place on WSB, consequently we zoom in on the market frenzy. We construct similar networks as before on a 5 minute frequency for the period 11 Jan 2021 until 13 Feb 2021. Note that the price peaked on 26 January, which marks the centre of this window. We apply the ISMB modelling approach to this

data using market variables constructed from minute-by-minute data. The results remain largely similar to the ones from the hour-by-hour analysis, with the notable exception that the effect of the change points of the VN and ANVN is less pronounced during the market frenzy than during the entire time period. We attribute this to the fact that during the frenzy, the market was a lot more volatile than before and actions taking place outside of WSB played a notable role for the market movements, such as a ban of Robinhood, the main trading app for WSB investors, on buying GME. Since the results support the findings made so far, we refer the interested reader to the Appendix 5 for a detailed analysis.

4.4 Reverse causality

Since we found that the change points of the VN and ANVN explain market movements, we also investigate if the opposite holds true, i.e. if the change points of the VN and ANVN result from the returns of GME. First, we regress the change points of the VN and ANVN with 7 Lags, covering the entire trading day, onto the present VN and ANVN variable realisation. By this we control for effects resulting from the change points itself. Then, we regress the returns of the past 7 lags onto the residuals of the former regression. The results are displayed in Table 10. We observe that none of the parameters is significant, except for the intercept. This infers that the change points of the VN and ANVN do not result from the returns on the markets, instead the cause for them originates in the network. Hence we follow that reverse causality is not present and the change points are not a result of changes in market returns.

4.5 Financial and economic implications

Our results show that there exists a link between individual users behaviour upon social networks and market behaviour. This provides not only empirical justification for the theoretical model by Pedersen (2022), but also shows the necessity to quantify individual users behaviour on social networks instead of aggregating over all users, as commonly done in many research studies. This necessity derives from the demonstrated relationship between individual users change in behaviour and market movements, which cannot be proxied for with aggregated network statistics.

The VNs take up a central role in this relationship, even though their own change in behaviour explains less market movements than the ones from ANVNs. VNs change in behaviour results after 6 hours in a significant effect upon market dynamics. This makes sense since a post by a single influential person first needs to reach a credible number of investors to lead to market reactions. While it is extremely challenging to detect change points in large social networks quickly (Yu et al., 2022; Zwetsloot et al., 2025), this suggests that influential users change in behaviour could act as an early warning sign for meme

Table 10: Regression to determine if the returns lead to the change points. First a regression onto the VN and ANVN variables with their past 7 lagged variables is run and the residuals from the regression are obtained. Then the past 7 returns are regressed upon the residuals of the former regression. We observe that the returns over 7 lags do not significantly impact the change point variables, hence the cause for the change points occurrence originates in the network.

	Dependent variable:	
	VN	ANVN
	(1)	(2)
ret1	−0.074	−0.067
	(0.141)	(0.141)
ret2	0.062	0.067
	(0.138)	(0.138)
ret3	−0.084	−0.081
	(0.138)	(0.138)
ret4	0.148	0.150
	(0.138)	(0.138)
ret5	−0.134	−0.129
	(0.138)	(0.138)
ret6	0.211	0.211
	(0.140)	(0.140)
ret7	−0.063	−0.063
	(0.139)	(0.139)
Constant	−0.00003	0.005***
	(0.002)	(0.002)
Observations	1,633	1,633
R^2	0.004	0.004
Note:	*p<0.1; **p<0.05; ***p<0.01	

stock market reactions, if identified early enough. We also find that several influential users exhibit a change in behaviour 1-2 months before the market frenzy. Lesser influential users (ANVNs) change in behaviour, if occurring in bulk, ripples instantly and with a various hours lasting effect into the financial markets. Their change in behaviour explains the largest portion of market dynamics, even when controlling for network activity. This finding paves the way towards the construction of variables for stock market movements based on social network users change in behaviour.

These findings emphasise the need to detect and monitor (change in behavior of) highly influential users, but it also shows the need to monitor their followership closely as it is their change in behaviour which explains an even larger part of market movements.

5 Discussion and Conclusion

In this study, we investigate the drivers and changes in users' posting probability on social networks via a sparse network model and change point detection modelling approach. For the study we collected 3.8 million posts from 502,650 users from WallStreetBets, a subreddit. We examine the impact of individual user behaviour on the Reddit forum Wallstreetbets upon markets. Thereby we provide empirical validation for the theoretical model developed by Pedersen (2022) as well as extending prior research that focuses on aggregated network measures only.

The DeGroot type model of Pedersen (2022) is a theoretical framework for understanding (meme) stock prices discussed and traded by users of social networks. It being a DeGroot model implies that users of the network change their opinions towards a stock as a result of information they receive from the network, which translates then into price changes. Our results demonstrate that changes in users' behaviour significantly explained returns, integrated volatility, and jump volatility, even when controlling for network activity and established metrics measuring influential user impact. We also show that the change in behaviour is not explained by market dynamics, instead it is rooted in the activity on the social network. Including changes in behaviour of users on Reddit into models to explain the market movements, leads to improvements in the modeling fit.

Studies often focus upon influential users in networks, but we show that changes in behaviour of less important users explain a larger part of returns, integrated variance and jump volatility than important users do. Influential users change in behaviour results after 6 hours in a significant effect upon market dynamics. This makes sense since a post by a single influential person first needs to reach a credible number of investors before eliciting market reactions. Lesser influential users change in behaviour, if occurring in bulk, ripples instantly and with a various hours lasting effect into the financial markets. Their change

in behaviour explains the largest portion of market dynamics, even when controlling for network activity. This finding paves the way towards the construction of variables for stock market movements based on social network users change in behaviour.

Our modelling approaches applicability extends well beyond the scope of the study. With the ever growing number of meme stocks, our method can be used to analyse the impact users on social media have upon market prices. While it is extremely challenging to detect change points quickly, our findings suggest that influential users change in behaviour could act as an early warning signal for meme stock market reactions, if identified early enough. We also find that several influential users exhibit a change in behaviour 1-2 months before the market frenzy. Further, our method is applicable to meme coins, which have similar origins as meme stocks. Due to the ease of creating new cryptocurrencies, the market of meme coins is persistently growing and our model provides a method to determine the effect social network users have upon meme coins.

This study sets the stage for a number of potential future research opportunities. We show in this study that individual users change in behaviour explains market movements. The GME market frenzy led to the emergence of various meme stocks, with some of them emerging shortly after the market frenzy, such as AMC and BB. A potential future research opportunity would be to study the spillover of the GME frenzy into these assets market frenzies and vice versa. In this study, we distinguish between the relationship of change in behaviour of vital and admissible non-vital users with market movements and show that the studies' results are robust towards different definitions for vital users. An interesting research question is the relationship between individual vital nodes and the admissible non-vital nodes associated with them, which would shed light onto the incremental role single users played in the market frenzy.

References

Agaskar, A. and Y. M. Lu (2013). "ALARM: A logistic auto-regressive model for binary processes on networks". In: *2013 IEEE Global Conference on Signal and Information Processing*. IEEE, pp. 305–308.

Akaike, H. (1974). "A new look at the statistical model identification". In: *IEEE transactions on automatic control* 19.6, pp. 716–723.

Almquist, Z. W. and C. T. Butts (2013). "Dynamic network logistic regression: A logistic choice analysis of inter-and intra-group blog citation dynamics in the 2004 US presidential election". In: *Political analysis* 21.4, pp. 430–448.

Aloosh, A., H.-E. Choi, and S. Ouzan (2021). "Meme stocks and herd behavior". In: *Available at SSRN 3909945*.

Andersen, T. G., T. Bollerslev, and F. X. Diebold (2007). "Roughing it up: Including jump components in the measurement, modeling, and forecasting of return volatility". In: *The review of economics and statistics* 89.4, pp. 701–720.

Andrei, D. and M. Hasler (2014). "Investor Attention and Stock Market Volatility". In: *The Review of Financial Studies* 28.1, pp. 33–72.

Aral, S. and D. Walker (2014). "Tie strength, embeddedness, and social influence: A large-scale networked experiment". In: *Management Science* 60.6, pp. 1352–1370.

Audrino, F., J. Chassot, C. Huang, M. Knaus, M. Lechner, and J.-P. Ortega (2024). "How does post-earnings announcement sentiment affect firms' dynamics? New evidence from causal machine learning". In: *Journal of Financial Econometrics* 22.3, pp. 575–604.

Audrino, F., F. Sigrist, and D. Ballinari (2020). "The impact of sentiment and attention measures on stock market volatility". In: *International Journal of Forecasting* 36.2, pp. 334–357.

Bai, P., A. Safikhani, and G. Michailidis (2023). "Multiple change point detection in reduced rank high dimensional vector autoregressive models". In: *Journal of the American Statistical Association* 118.544, pp. 2776–2792.

Bailey, M., D. Johnston, M. Koenen, T. Kuchler, D. Russel, and J. Stroebel (2024). "Social networks shape beliefs and behavior: Evidence from social distancing during the covid-19 pandemic". In: *Journal of Political Economy Microeconomics* 2.3, pp. 463–494.

Bamakan, S. M. H., I. Nurgaliev, and Q. Qu (2019). "Opinion leader detection: A methodological review". In: *Expert Systems with Applications* 115, pp. 200–222.

Barigozzi, M. and C. Brownlees (2019). "NETS: Network estimation for time series". In: *Journal of Applied Econometrics* 34.3, pp. 347–364.

Barros, C. D., M. R. Mendonça, A. B. Vieira, and A. Ziviani (2021). "A survey on embedding dynamic graphs". In: *ACM Computing Surveys (CSUR)* 55.1, pp. 1–37.

Bartov, E., L. Faurel, and P. Mohanram (2023). "The role of social media in the corporate bond market: Evidence from Twitter". In: *Management Science* 69.9, pp. 5638–5667.

Behrendt, S. and A. Schmidt (2018). "The Twitter myth revisited: Intraday investor sentiment, Twitter activity and individual-level stock return volatility". In: *Journal of Banking & Finance* 96, pp. 355–367.

Beladev, M., G. Katz, L. Rokach, U. Singer, and K. Radinsky (2023). "GraphERT–Transformers-based Temporal Dynamic Graph Embedding". In: *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, pp. 68–77.

Belloni, A. and V. Chernozhukov (2013). "Least squares after model selection in high-dimensional sparse models". In: *Bernoulli* 19.2, pp. 521–547.

Bhattacharya, P., T. Q. Phan, X. Bai, and E. M. Airoldi (2019). "A coevolution model of network structure and user behavior: The case of content generation in online social networks". In: *Information Systems Research* 30.1, pp. 117–132.

Chen, C. Y.-H., W. K. Härdle, and Y. Klochkov (2022). "Sonic: Social network analysis with influencers and communities". In: *Journal of Econometrics* 228.2, pp. 177–220.

Chen, C. Y.-H., Y. Okhrin, and T. Wang (2024). "Monitoring network changes in social media". In: *Journal of Business & Economic Statistics* 42.2, pp. 391–406.

Chen, E. Y., J. Fan, and X. Zhu (2023). "Community network auto-regression for high-dimensional time series". In: *Journal of Econometrics* 235.2, pp. 1239–1256.

Costola, M., M. Iacopini, and C. R. Santagiustina (2021). "On the 'mementum' of meme stocks". In: *Economics Letters* 207, p. 110021.

de Paula, Á. (2017). "Econometrics of Network Models". In: *Advances in Economics and Econometrics: Eleventh World Congress*. Ed. by B. Honoré, A. Pakes, M. Piazzesi, and L. Samuelson. Cambridge University Press, pp. 268–323.

DeGroot, M. H. (1974). "Reaching a consensus". In: *Journal of the American Statistical association* 69.345, pp. 118–121.

Deng, J., M. Yang, M. Pelster, and Y. Tan (2024). "Social trading, communication, and networks". In: *Information Systems Research* 35.4, pp. 1546–1564.

Fan, J. and R. Li (2001). "Variable Selection via Nonconcave Penalized Likelihood and its Oracle Properties". In: *Journal of the American Statistical Association* 96.456, pp. 1348–1360.

GameStop Stock Price Timeline (2024). `https://gmetimeline.com/`. Accessed: 12 April 2024.

Gan, B., V. Alexeev, R. Bird, and D. Yeung (2020). "Sensitivity to sentiment: News vs social media". In: *International Review of Financial Analysis* 67, p. 101390.

Gelper, S., R. van der Lans, and G. van Bruggen (2021). "Competition for attention in online social networks: Implications for seeding strategies". In: *Management Science* 67.2, pp. 1026–1047.

Ghezzi, F., E. Rossi, and L. Trapani (2024). "Fast online changepoint detection". In: *arXiv preprint arXiv:2402.04433*.

Gopal, R. D., X. Qiao, M. S. Strub, and Z. Yang (2024). "Gaining a seat at the table: Enhancing the attractiveness of online lending for institutional investors". In: *Information Systems Research*.

Gu, C. and A. Kurov (2020). "Informational role of social media: Evidence from Twitter sentiment". In: *Journal of Banking & Finance* 121, p. 105969.

Han, Y., T. Lappas, and G. Sabnis (2020). "The importance of interactions between content characteristics and creator characteristics for studying virality in social media". In: *Information Systems Research* 31.2, pp. 576–588.

He, Q., Y. Hong, and T. Raghu (2024). "Platform governance with algorithm-based content moderation: An empirical study on Reddit". In: *Information Systems Research*.

Hui, Y., I. M. Zwetsloot, S. Trimborn, and S. Rudinac (2024). "Domain-Informed Negative Sampling Strategies for Dynamic Graph Embedding in Meme Stock-Related Social Networks". In: *arXiv preprint arXiv:2411.00606*.

Ji, L., X. Chen, L. Jiang, and J. Huang (2025). "Within-and Cross-group Spillover Effects in Influencer Marketing: The Heterogeneity Between Micro-and Macro-influencers". In: *Information & Management*, p. 104202.

Jiao, P., A. Veiga, and A. Walther (2020). "Social media, news media and the stock market". In: *Journal of Economic Behavior & Organization* 176, pp. 63–90.

Kendrick, L., K. Musial, and B. Gabrys (2018). "Change point detection in social networks—Critical review with experiments". In: *Computer Science Review* 29, pp. 1–13.

Kim, B., K. H. Lee, L. Xue, and X. Niu (2018). "A review of dynamic network models with latent variables". In: *Statistics surveys* 12, p. 105.

Klein, T. (2022). "A note on GameStop, short squeezes, and autodidactic herding: An evolution in financial literacy?" In: *Finance Research Letters* 46, p. 102229.

Lehrer, S., T. Xie, and T. Zeng (2021). "Does high-frequency social media data improve forecasts of low-frequency consumer confidence measures?" In: *Journal of Financial Econometrics* 19.5, pp. 910–933.

Li, F. and T. C. Du (2011). "Who is talking? An ontology-based opinion leader identification framework for word-of-mouth marketing in online social blogs". In: *Decision support systems* 51.1, pp. 190–197.

Li, J. and Z. Li (2024). "Sentiment, Social Media, and Meme Stock Return Predictability". In: *Social Media, and Meme Stock Return Predictability (June 21, 2024)*.

Lu, Y., K. Jerath, and P. V. Singh (2013). "The emergence of opinion leaders in a networked online community: A dyadic model with time dynamics and a heuristic for fast estimation". In: *Management Science* 59.8, pp. 1783–1799.

Lü, L., D. Chen, X.-L. Ren, Q.-M. Zhang, Y.-C. Zhang, and T. Zhou (2016). "Vital nodes identification in complex networks". In: *Physics Reports* 650, pp. 1–63.

Mallik, A. and Z. W. Almquist (2019). "Stable Multiple Time Step Simulation/Prediction From Lagged Dynamic Network Regression Models". In: *Journal of Computational and Graphical Statistics* 28.4, pp. 967–979.

Mölders, M., L. Bock, E. Barrantes, and H. Zülch (2025). "Understanding finfluencers: Roles and strategic partnerships in retail investor engagement". In: *Journal of Business Research* 198, p. 115462.

Morone, F. and H. A. Makse (2015). "Influence maximization in complex networks through optimal percolation". In: *Nature* 524.7563, pp. 65–68.

Niu, Y. S., N. Hao, and H. Zhang (2016). "Multiple change-point detection: a selective overview". In: *Statistical Science*, pp. 611–623.

Pedersen, L. H. (2022). "Game on: Social networks and markets". In: *Journal of Financial Economics* 146.3, pp. 1097–1119.

Peng, L. and L. Zhang (2024). "Unleashing the crowd: the effect of social networks in crowdfunding markets". In: *Management Science*.

SEC (2021). *Staff report on equity and options market structure conditions in early 2021*. Tech. rep. US Securities and Exchange Commission.

Tibshirani, R. (1996). "Regression Shrinkage and Selection via the Lasso". In: *Journal of the Royal Statistical Society. Series B (Methodological)* 58.1, pp. 267–288.

Trimborn, S., Y. Chen, and R.-B. Chen (2022). "Influencers, inefficiency and fraud-The Bitcoin price discovery network under the microscope". In: *Available at SSRN*.

Trimborn, S., H. Peng, and Y. Chen (2024). "Influencer Detection Meets Network AutoRegression–Influential Regions in the Bitcoin Blockchain". In: *Journal of Empirical Finance* 78, p. 101529.

Umar, Z., M. Gubareva, I. Yousaf, and S. Ali (2021). "A tale of company fundamentals vs sentiment driven pricing: The case of GameStop". In: *Journal of Behavioral and Experimental Finance* 30, p. 100501.

Vuong, Q. H. (1989). "Likelihood ratio tests for model selection and non-nested hypotheses". In: *Econometrica: journal of the Econometric Society*, pp. 307–333.

Wang, Z., S. Hao, I. M. Zwetsloot, and S. Trimborn (2024). "Social Network Datasets on Reddit Financial Discussion". In: *arXiv preprint arXiv:2410.05002*.

Woodall, W. H., M. J. Zhao, K. Paynabar, R. Sparks, and J. D. Wilson (2017). "An overview and perspective on social network monitoring". In: *IISE Transactions* 49.3, pp. 354–365.

Yang, S. Y., S. Y. K. Mo, and A. Liu (2015). "Twitter financial community sentiment and its predictive relationship to stock market movement". In: *Quantitative Finance* 15.10, pp. 1637–1656.

Yon, G. G. V., A. Slaughter, and K. de la Haye (2021). "Exponential random graph models for little networks". In: *Social Networks* 64, pp. 225–238.

Yousaf, I., L. Pham, and J. W. Goodell (2023). "The connectedness between meme tokens, meme stocks, and other asset classes: Evidence from a quantile connectedness approach". In: *Journal of International Financial Markets, Institutions and Money* 82, p. 101694.

Yu, L., L. Sun, B. Du, and W. Lv (2023). "Towards better dynamic graph learning: New architecture and unified library". In: *Advances in Neural Information Processing Systems* 36, pp. 67686–67700.

Yu, L., I. M. Zwetsloot, N. T. Stevens, J. D. Wilson, and K. L. Tsui (2022). "Monitoring dynamic networks: A simulation-based strategy for comparing monitoring methods and a comparative study". In: *Quality and Reliability Engineering International* 38.3, pp. 1226–1250.

Zhang, B., P. A. Pavlou, and R. Krishnan (2018). "On direct vs. indirect peer influence in large social networks". In: *Information Systems Research* 29.2, pp. 292–314.

Zhang, K. and S. Trimborn (2023). "Influential assets in Large-Scale Vector AutoRegressive Models". In: *SSRN Working paper*.

Zou, H. (2006). "The Adaptive Lasso and Its Oracle Properties". In: *Journal of the American Statistical Association* 101.476, pp. 1418–1429.

Zwetsloot, I. M., N. T. Stevens, and Y. Hui (2025). "Online Change Detection in Large-Scale Dynamic Networks". In: *Working paper*.

Online Appendix

A Robustness analysis: Vital Node definition

In Definition 1 we postulate a definition for vital nodes as *those nodes in the network that submit posts in at least 10% of the weeks and receive the highest 10% median comments on their posts.*

Figure A.1 shows the network structure of the 64 vital nodes. The node size represents the number of ANVN for that vital node. The edges indicate if a vital node is an ANVN of the other vital node and the edge weights represent number of communications between the two. It is clear that a large number of vital nodes also follow each other and that there is a large variety in the number of ANVNs per vital node.

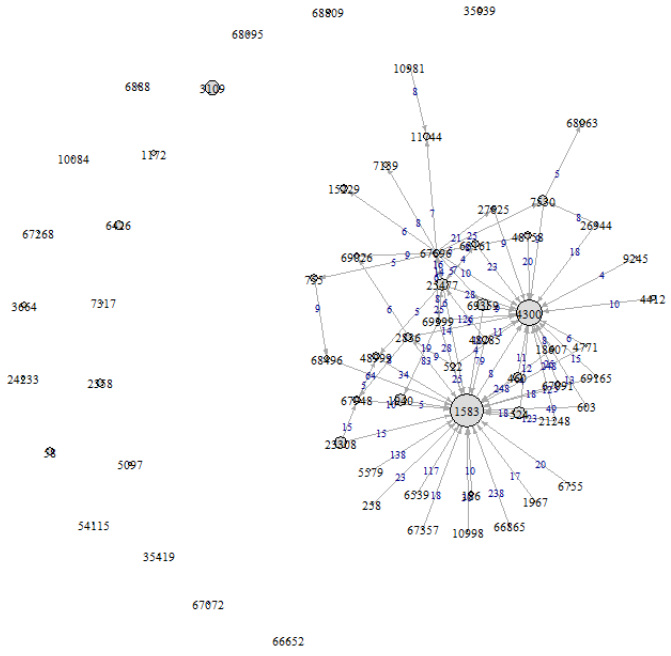
Definition 1 is a relaxed definition as it only requires posting activity in 5 weeks. To validate if a stricter definition of a vital node would impact the results of the study, we investigate the robustness of the results subject to the definition by rerunning the full analysis while adapting this definition.

We amend the VN definition by the posting frequency and amount of comments. We redefine a VN as communicating in at least 15% and 20% of the weeks (i.e. more frequently). In addition, the 10% median comments is also validated by requiring the highest 5% median comments (making it stricter). We perform the robustness analysis for each combination of the stated values. First, we rerun our algorithm to identify all nodes that qualify as VNs and as ANVNs according to these stricter definitions. Table A.1 shows for each of these values how many VNs and ANVNs are identified in the network. The row 'Original' shows the values as used Definition 1 which resulted in 64 VNs and 48211 ANVNs. As we become stricter in our definition we see that the number of identified VNs declines, the number of ANVNs only slightly declines. This implies already that large parts of the communications are focused upon a few highly influential nodes.

Table A.1: VN identified when changing Definition 1 (in bold the definition used in the main analysis)

Scenario	% of weeks	% median comments	No. of VNs	No. of ANVNs
Robustness: 10-5	10	5	47	48149
Robustness: 15-5	15	5	14	48018
Robustness: 20-5	20	5	9	47993
Original: 10-10	10	10	64	48211
Robustness: 15-10	15	10	16	48021
Robustness: 20-10	20	10	9	47993

Figure A.1: This Figure shows the network structure of the 64 vital nodes. The node size represents the number of ANVNs for each vital node. The edges indicate if a vital node is an ANVN of the other vital node and the edge weights represent number of communications between the two. It is clear that a large number of vital nodes also follow each other and that there is a large variety in the number of ANVNs per vital node.



As we modify the VN definition, the estimated change point sequences (Figure 8) will also alter. We obtain these new change point sequences (equations 5 and 6) and rerun the analysis of linking the change points to the market as modeled in Equation 7. The R^2 and adj. R^2 of these rerun models are reported in Tables A.2, A.3, and A.4. Each row indicates a different definition for the VN identification and the resulting ANVNs (as shown in Table A.1). Hence, we validate if a change in the VN definition and as a result a change in the number of identified change points, affects the main results of this study.

Tables A.2, A.3, and A.4 show similar results than reported in section 5 (highlighted in bold in each Table). Regardless of the definition for the vital nodes, we observe that the identified change points strongly improve the modelling of the return, integrated variance, and jump volatility. Notably, we observe that for several instances, the base setting of this study, '10-10', leads to a lower R^2 and adj. R^2 than other settings. This result let us to using '10-10' as the base setting for this study. Hence, we conclude that the results of the study are robust towards the VN definition and that the chosen definition is a conservative choice.

Table A.2: Return robustness: This table shows that the results are robust towards our vital node definition made in Definition 1. Displayed are the R^2 and adj. R^2 for model 7. Column (1) shows the results for the autoregressive benchmark model, (2)-(6) add variables to the benchmark model. We analyse if the results are robust towards weaker/stricter requirements for submitting a post in 10, 15, 20 % of the weeks and highest 5, 10 % median comments. Note that 'Original: 10 - 10' corresponds to the definition in the main analysis of the paper. The results show that regardless of the definition of a vital node, the change points of the VN and ANVN strongly improve the modelling of the return series.

		(1)	(2)	(3)	(4)	(5)	(6)
Robustness: 10 - 5	R^2	0.009	0.02	0.03	0.091	0.045	0.118
	adj. R^2	0.004	0.011	0.016	0.074	0.021	0.088
Robustness: 15 - 5	R^2	0.009	0.02	0.04	0.097	0.045	0.125
	adj. R^2	0.004	0.011	0.026	0.08	0.021	0.095
Robustness: 20 - 5	R^2	0.009	0.02	0.049	0.106	0.045	0.135
	adj. R^2	0.004	0.011	0.036	0.089	0.021	0.105
Original: 10 - 10	R^2	0.009	0.02	0.028	0.088	0.045	0.115
	adj. R^2	0.004	0.011	0.014	0.07	0.021	0.084
Robustness: 15 - 10	R^2	0.009	0.02	0.035	0.093	0.045	0.12
	adj. R^2	0.004	0.011	0.022	0.075	0.021	0.089
Robustness: 20 - 10	R^2	0.009	0.02	0.049	0.106	0.045	0.135
	adj. R^2	0.004	0.011	0.036	0.089	0.021	0.105
	Vital nodes	No	No	Yes	Yes	No	Yes
	Non-vital nodes	No	No	No	Yes	No	Yes
	Network activity	No	Yes	Yes	Yes	Yes	Yes
	Network statistics	No	No	No	No	Yes	Yes

B Change point detection details

We specify a change point detection framework, to link individual change behaviour to the stock market behavior. The method is based upon a greedy algorithm for searching for

Table A.3: Integrated variance robustness: This table shows that the results are robust towards our vital node definition made in Definition 1. Displayed are the R^2 and adj. R^2 for model 7. Column (1) shows the results for the autoregressive benchmark model, (2)-(6) add variables to the benchmark model. We analyse if the results are robust towards weaker/stricter requirements for submitting a post in 10, 15, 20 % of the weeks and highest 5, 10 % median comments. Note that 'Original: 10 - 10' corresponds to the definition in the main analysis of the paper. The results show that regardless of the definition of a vital node, the change points of the VN and ANVN strongly improve the modelling of the return series.

		(1)	(2)	(3)	(4)	(5)	(6)
Robustness: 10 - 5	R^2	0.175	0.189	0.256	0.288	0.255	0.336
	adj. R^2	0.172	0.182	0.245	0.274	0.237	0.313
Robustness: 15 - 5	R^2	0.175	0.189	0.19	0.21	0.255	0.269
	adj. R^2	0.172	0.182	0.179	0.195	0.237	0.244
Robustness: 20 - 5	R^2	0.175	0.189	0.19	0.211	0.255	0.27
	adj. R^2	0.172	0.182	0.179	0.196	0.237	0.245
Original: 10 - 10	R^2	0.175	0.189	0.239	0.268	0.255	0.317
	adj. R^2	0.172	0.182	0.228	0.254	0.237	0.293
Robustness: 15 - 10	R^2	0.175	0.189	0.19	0.21	0.255	0.269
	adj. R^2	0.172	0.182	0.179	0.195	0.237	0.244
Robustness: 20 - 10	R^2	0.175	0.189	0.19	0.211	0.255	0.27
	adj. R^2	0.172	0.182	0.179	0.196	0.237	0.245
	Vital nodes	No	No	Yes	Yes	No	Yes
	Non-vital nodes	No	No	No	Yes	No	Yes
	Network activity	No	Yes	Yes	Yes	Yes	Yes
	Network statistics	No	No	No	No	Yes	Yes

multiple change points in time series data, cf. Bai et al. (2023). Our implementation of this general framework is visualized in Figure 3. The algorithm we propose consists of three steps (A-C), is entirely data driven, and is implemented as an exhaustive search algorithm:

Step A: Identify change point candidates by conducting a rolling window search and identifying a single (possible) change point in each window.

Step B: Obtain all possible change point sequences by constructing a set of unique change points and building the powerset.

Step C: Select the best change point sequence by estimating a model for each combination of change points and selecting the best model and related change point sequence.

Next, we discuss each step in detail.

Step A: Identify change point candidates

In Step A, we conduct a rolling window search over windows of length h. Between the starting points of 2 windows, we define a step size $l \geq 1$. For a given dataset of length T, there are G windows of length h with step size l. For $g \in \{1, ..., G\}$, define $T_{1,g}$, $T_{2,g}$ as the start and endpoint of window g respectively. Next, we wish to derive the most likely

Table A.4: Jump volatility robustness: This table shows that the results are robust towards our vital node definition made in Definition 1. Displayed are the R^2 and adj. R^2 for model 7. Column (2) shows the results for the benchmark model containing the network activity as variables, (3)-(6) add variables to the benchmark model. We analyse if the results are robust towards weaker/stricter requirements for submitting a post in 10, 15, 20 % of the weeks and highest 5, 10 % median comments. Note that 'Original: 10 - 10' corresponds to the definition in the main analysis of the paper. The results show that regardless of the definition of a vital node, the change points of the VN and ANVN strongly improve the modelling of the return series.

		(2)	(3)	(4)	(5)	(6)
Robustness: 10 - 5	R^2	0.022	0.034	0.056	0.102	0.133
	adj. R^2	0.017	0.025	0.042	0.084	0.107
Robustness: 15 - 5	R^2	0.022	0.022	0.044	0.102	0.123
	adj. R^2	0.017	0.012	0.03	0.084	0.096
Robustness: 20 - 5	R^2	0.022	0.022	0.044	0.102	0.123
	adj. R^2	0.017	0.013	0.03	0.084	0.097
Original: 10 - 10	R^2	0.022	0.032	0.054	0.102	0.132
	adj. R^2	0.017	0.022	0.04	0.084	0.106
Robustness: 15 - 10	R^2	0.022	0.022	0.044	0.102	0.123
	adj. R^2	0.017	0.012	0.03	0.084	0.096
Robustness: 20 - 10	R^2	0.022	0.022	0.044	0.102	0.123
	adj. R^2	0.017	0.013	0.03	0.084	0.097
	Vital nodes	No	Yes	Yes	No	Yes
	Non-vital nodes	No	No	Yes	No	Yes
	Network activity	Yes	Yes	Yes	Yes	Yes
	Network statistics	No	No	No	Yes	Yes

change point within each window g. We define the set of possible change points in window G as T_G, spanning every datapoint in the inner 50% quantile range. By this, it is ensured sufficient data before and after each change point exist for model estimation. For each possible change point τ in window g we estimate equation (2) or (3) by minimizing the function $L()$ which is the difference between realised and predicted values:

$$L(T_{1,g}, \tau) = \frac{1}{\tau - T_{1,g} + 1} \arg\min_{A} \sum_{t=T_{1,g}}^{\tau} (y_t - I(\text{logit}(AX_t) > 0.5)) \tag{B.1}$$

$$L(\tau + 1, T_{2,g}) = \frac{1}{T_{2,g} - \tau} \arg\min_{A} \sum_{t=\tau+1}^{T_{2,g}} (y_t - I(\text{logit}(AX_t) > 0.5)), \tag{B.2}$$

where $L(a, b)$ is related to the model loss based on the data from timepoint a to b. Then, the final change point candidate in window g is the minimizer of

$$\hat{\tau}_g = \arg\min_{\tau} L(T_{1,g}, \tau) + L(\tau + 1, T_{2,g}).$$

Step B: Identify possible change point sequences

We derive the change point candidates for all windows G, which provides us with a set of $\{\hat{\tau}_g\}_1^G$. Define the set S_τ as the set with all unique elements in $\{\hat{\tau}_g\}_1^G$. Next, we obtain $P(S_\tau)$, the power set of S_τ containing all subsets of S_τ, including the empty set and S_τ itself.

Step C: Select the best change point sequence

Take an element $CPs \in P(S_\tau)$, where CPs is a sequence of change points: i.e. $CPs = \{cp_1, \ldots, cp_R\}$, where $R = |CPs|$ is the number of change points, a non-negative integer. Note that CPs cuts the total dataset into $R+1$ consecutive windows. We denote by T_{1,cp_r} and T_{2,cp_r} the start and endpoint of window r with $r \in \{1, \ldots R\}$. We estimate the model for the full dataset of length T with change point set CPs and compute $L(CPs)$, which is obtained as

$$L(CPs) = \sum_{r=1}^{R} L(T_{1,r}, T_{2,r}) + \omega n_{CPs}, \tag{B.3}$$

as in Bai et al. (2023). To control for the likelihood improvement when additional change points are included, we incorporate ωn_{CPs} where ω is the penalty parameter and n_{CPs} is the number of non-zero parameters estimated in the $R+1$ models (i.e. the number of non-zero elements in Θ_s in model (4), summed up over the $R+1$ windows). As we are interested in a model with good forecasting accuracy, we choose to work with the AIC criterion, hence $\omega = 2$ (Akaike, 1974). We repeat this procedure for each element of CPs in $P(S_\tau)$. Note that for the empty element of CPs, i.e. no change points in the model, equation (B.3) simplifies to $L(CPs = \emptyset) = L(1, T) + \omega n_{CPs}$. The final change point set for this model is the minimizer of

$$\widehat{CPs} = \underset{CPs \in P(S_\tau)}{\arg\min} \; (-2\, L(CPs)). \tag{B.4}$$

This results in a sequence of change points $\widehat{CPs}$ that minimizes the penalized loss over the whole dataset. These change points determine the time points the communication behaviour between the nodes i and j changed.

To ensure that the final change point sequence is not only outperforming in terms of likelihood but also does so with statistical significance, we perform the Vuong test for overlapping models (Vuong, 1989) on a 5% level, which is akin to a likelihood ratio test. We test the models which minimize (B.4) against a model without any change points. We perform the Vuong test for overlapping models as the models potentially overlap. The test statistic is compared against critical values from the sum of weighted chi-squared random

variables, which differentiates the test from the classical likelihood ratio test. As a result, we only consider series of changes in communication behaviour (change points series) which are statistically significantly better on a 5% level compared to the model with no changes in behaviour.

Model selection and interpretability

We use sparse modelling methods (Barigozzi and Brownlees, 2019; Trimborn et al., 2022; Trimborn et al., 2024; Zhang and Trimborn, 2023) to identify the relevant determinants of communication between two nodes in a data-driven manner. The models as defined above include many parameters. For example, assume we identified R change point candidates, than we have $R+1$ submodels. Each submodel has either 32 or 34 features as listed in Table 3 and these features are included for K lags. This results in a model with hundreds of parameters, challenging the interpretability of the parameters. To tackle this issue, we perform LASSO regularization (Tibshirani, 1996), which is an often applied technique in network analysis (Barigozzi and Brownlees, 2019; Trimborn et al., 2022; Trimborn et al., 2024; Zhang and Trimborn, 2023). LASSO enables us to detect which parameters are relevant for the users posting behaviour. After we identified the relevant parameters, we estimate the equations (2) and (3) with the change points $\widehat{CPs}$ as estimated in Eq (B.4) again but only including the parameters selected in the previous step. This step follows the finding of Belloni and Chernozhukov (2013) that a reestimation of the model removes the bias introduced by the LASSO estimation procedure.

C Speeding up computational time

The ISBM method presented in this study is a very greedy algorithm and therefore is time consuming to run. We incorporate the following strategies to speed up computation:

- In step A, the algorithm computes the likelihood over the window for each possible change point in the window. To speed up this step, we first compute the likelihood for each 12th change point in the window using the change point which yields maximum likelihood. Then we compute the likelihood for the change point in all 11 steps before and after the identified change point and use the change point that gives the maximum value as the final result.
- The set S_τ often consists of many possible change points resulting in $P(S_\tau)$ having a large number of elements. To prevent overfitting and to speed up computation, we exclude all elements of 5 or more change points from $P(S_\tau)$. Our final results (see

Table D.1: Identified change points

No. cps	VN	%	ANVN	%
0	38	59.4	2895	13.5
1	26	40.6	18566	86.4
2	0	0.0	16	0.1
3	0	0.0	1	0.0
≥ 4	0	0.0	0	0.0

section 5) show that the maximum number of identified change points is 3, so this does not seem a restrictive strategy.

- The set S_τ often consists of many possible change points that are close together. We therefore restrict the set $P(S_\tau)$ to have only elements for which consecutive change points are at least k time units apart. From an interpretation point of view it seems reasonable to assume that a Reddit user does not change his/her behaviour too frequently. In addition, estimating models for sparse data in such short time spans is challenging.

D Robustness analysis: 5-minute-by-5-minute model

Naturally the period of the market frenzy is of high interest. Due to the speed at which things changed during that time, we analyse it at a 5-minute frequency. Similarly to the hourly setting, we apply the models (2) and (3) and combine the models with the change point detection method described in section 4.2. We perform the change point selection under the settings window size: $h = 2016$ 5-minute intervals (1 week), stepsize: $l = 288$ 5-minute intervals (1 day), and time between change points: $k = 288$ (1 day). We analyse the network around the frenzy (11 Jan 2021 until 13 Feb 2021).

We first look at the identified change points in our models. Table D.1 displays the results, and we see that most VNs, 59.4%, do not change their behaviour in the period around the frenzy but that many of the ANVNs, 86.4%, do show a change in posting behaviour. Figure D.1 shows the change points of all VNs (left) and all ANVNs (right) versus time. We see that the VN change points are spread out over the month of January whereas most ANVNs change their behaviour towards the end of it.

Next, we are going to analyse the parameters of the models. As discussed in section 4.2.1 we use LASSO to ensure interpretability of the models we estimate. For the VNs only two vital nodes have non-zero parameters in the estimated models (excluding a non-zero constant), the first VN is 4300 (Keith Gill) and the other is VN 24233. The parameters of VN 4300 are summarized in Table D.2. We see that past own activity reduces likelihood

Figure D.1: Changepoints over time for the VNs (left) and the ANVNs (right).

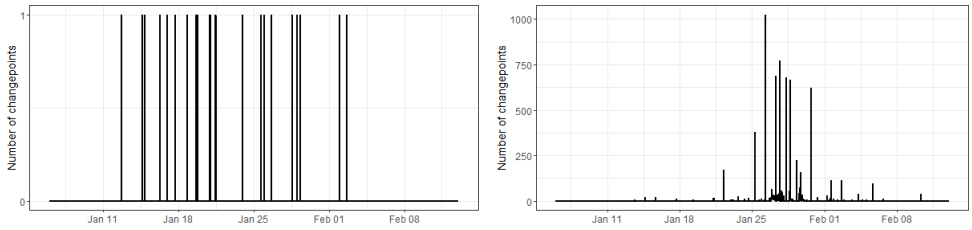


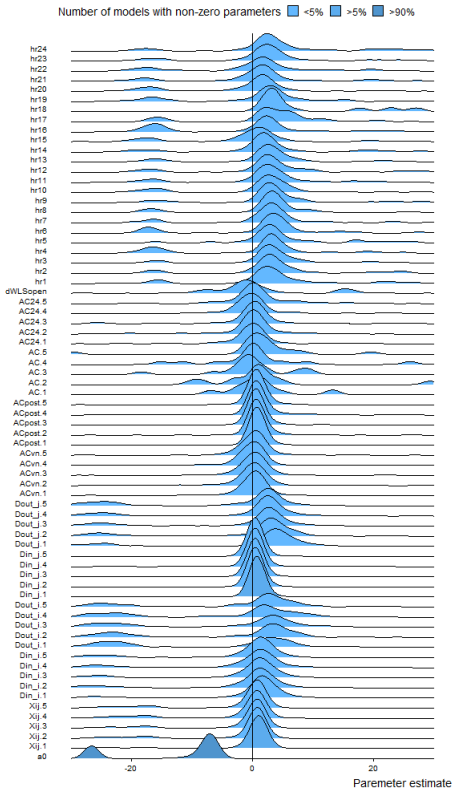
Table D.2: Parameters for VN 4300 in 5 minute model

Variable	Lag	Parameter estimate
a0		-46.9
x_{ii}	1	-36.8
x_{ii}	2	-54.0
x_{ii}	3	-51.7
x_{ii}	4	-46.0
x_{ii}	5	-29.5
D_i^{in}	1	1.9
D_i^{in}	2	3.6
D_i^{in}	4	-2.6
D_i^{in}	5	-3.5
AC_1^{vn}	1	1.4
AC_1^{vn}	3	0.2
AC_1^{vn}	5	-0.8
AC	1	-0.5
AC	5	0.9
AC^{24}	1	-0.002
AC^{24}	2	-0.6
AC^{24}	3	0.5
Dum_{wls}		18.0
hr21		23.7

of posting in addition to when Wallstreet is open and at hour 21 (just after closing of Wallstreet in UTC time) the likelihood of posting is increased. All other parameters are rather close to zero.

Figure D.2 shows the parameter estimates for the ANVN models and the pattern is very similar to the hourly model (recall Figure 5, right Figure). The only large difference is that the parameters estimated for the overall activity (AC) are more dispersed, this is unexpected as the overall activity level in the frenzy period is also much higher and varies more compared to the period before the frenzy (refer to Figure 1). For the impact of the change in behaviour on the markets during the frenzy, see Appendix E.

Figure D.2: Parameter Estimates 5-minute model for ANVN



E WSB impact during the market frenzy: 5-minute-by-5-minute model

We compute the 5-minute RV from the minute-by-minute returns of GME and construct the IV and JV by the Andersen et al. (2007) method, similar as for the hourly data. We estimate the models with 12 lags, representing the lagged observations of the past hour, and summarise the results in Tables E.1, E.2 and E.3for return, IV and JV respectively. We observe that adding the VN and ANVN to the model increases the adjusted R^2 strongly. As for IV, considering both VN and ANVN increases the adj. R^2 to 0.217 whereas a model consisting only of lagged variables spanning the past IV's, provides an adj. R^2 of 0.179. For the VN we observe that the 5 and 20 minute lagged ones are significant, hence the impact from a change in behaviour of the vital nodes during the market frenzy affects the integrated volatility of GME immediately. For the ANVNs a change in their behaviour transmits slower to the integrated variance as the 35, 55 and 60 minute lagged ones are significant. Next, we analysed if conventional network statistics proxy the explained movements from the VN and ANVN. The adjusted R^2 is much lower and renders it worse than the models containing only the VN and ANVN's. Considering all variables jointly results in an adjusted R^2 lower than the one for the model not containing the network statistics. This suggests that the VN

Table E.1: Significant variables from models with differing variables include and Lag length 12. Insignificant variables are not displayed. Control variables (network statistics and network activity) are included for differing models.

	Dependent variable:					
	ret					
	(1)	(2)	(3)	(4)	(5)	(6)
ret2	−0.145***	−0.150***	−0.155***	−0.132***	−0.153***	−0.134***
	(0.047)	(0.047)	(0.047)	(0.049)	(0.048)	(0.049)
ret3	−0.260***	−0.251***	−0.251***	−0.217***	−0.257***	−0.224***
	(0.047)	(0.047)	(0.048)	(0.050)	(0.048)	(0.051)
ret4	−0.180***	−0.176***	−0.174***	−0.158***	−0.174***	−0.158***
	(0.053)	(0.053)	(0.053)	(0.054)	(0.054)	(0.055)
ret5	−0.038	−0.044*	−0.046*	−0.042	−0.048*	−0.045*
	(0.026)	(0.026)	(0.027)	(0.027)	(0.027)	(0.027)
ret6	0.047	0.056	0.057	0.078*	0.062	0.084*
	(0.043)	(0.043)	(0.044)	(0.045)	(0.044)	(0.045)
ret10	0.073***	0.070***	0.071**	0.063**	0.065**	0.059**
	(0.027)	(0.027)	(0.027)	(0.028)	(0.028)	(0.028)
ANVN7				−0.0001**		−0.0001**
				(0.00003)		(0.00003)
Constant	0.002*	0.0003	0.0003	0.001	−0.018	−0.020
	(0.001)	(0.002)	(0.002)	(0.002)	(0.021)	(0.021)
Vital nodes	No	No	Yes	Yes	No	Yes
Non-vital nodes	No	No	No	Yes	No	Yes
Network activity	No	Yes	Yes	Yes	Yes	Yes
Network statistics	No	No	No	No	Yes	Yes
Observations	1,582	1,582	1,582	1,582	1,582	1,582
R^2	0.057	0.066	0.069	0.078	0.075	0.087
Adjusted R^2	0.050	0.051	0.046	0.047	0.036	0.032
Note:						*p<0.1; **p<0.05; ***p<0.01

Table E.2: Significant variables from models with differing variables include and Lag length 7. Insignificant variables are not displayed. Control variables (network statistics and network activity) are included for differing models.

	Dependent variable:					
	IV					
	(1)	(2)	(3)	(4)	(5)	(6)
IV1	0.115***	0.101***	0.105***	0.105***	0.087***	0.094***
	(0.025)	(0.025)	(0.025)	(0.025)	(0.025)	(0.025)
IV2	0.161***	0.147***	0.136***	0.135***	0.135***	0.125***
	(0.026)	(0.026)	(0.027)	(0.027)	(0.027)	(0.027)
IV3	0.097***	0.081***	0.088***	0.081***	0.075***	0.076***
	(0.025)	(0.025)	(0.026)	(0.026)	(0.026)	(0.026)
IV4	0.167***	0.154***	0.149***	0.152***	0.145***	0.144***
	(0.026)	(0.026)	(0.027)	(0.027)	(0.027)	(0.027)
IV10	0.058**	0.060**	0.047*	0.052**	0.055**	0.050*
	(0.025)	(0.025)	(0.025)	(0.025)	(0.025)	(0.026)
IV11	0.060**	0.064**	0.064**	0.082***	0.060**	0.078***
	(0.026)	(0.026)	(0.026)	(0.027)	(0.027)	(0.027)
IV12	0.073***	0.073***	0.081***	0.050**	0.070***	0.048*
	(0.024)	(0.024)	(0.024)	(0.025)	(0.025)	(0.025)
VN1			0.002**	0.002**		0.002**
			(0.001)	(0.001)		(0.001)
VN4			0.004***	0.004***		0.004***
			(0.001)	(0.001)		(0.001)
VN10			−0.0003	−0.0002		−0.0002
			(0.001)	(0.001)		(0.001)
VN11			−0.0002	−0.0003		−0.0003
			(0.001)	(0.001)		(0.001)
VN12			−0.0002	−0.0001		−0.0002
			(0.001)	(0.001)		(0.001)
ANVN1				−0.00000		−0.00000
				(0.00000)		(0.00000)
ANVN4				−0.00000		−0.00000
				(0.00000)		(0.00000)
ANVN7				0.00001***		0.00001***
				(0.00000)		(0.00000)
ANVN10				−0.00000		−0.00000
				(0.00000)		(0.00000)
ANVN11				0.00003***		0.00003***
				(0.00001)		(0.00001)
ANVN12				0.00001**		0.00001**
				(0.00000)		(0.00000)
Constant	0.0003***	−0.0001	−0.0001	−0.0001	0.001	0.001
	(0.0001)	(0.0002)	(0.0002)	(0.0002)	(0.002)	(0.002)
Vital nodes	No	No	Yes	Yes	No	Yes
Non-vital nodes	No	No	No	Yes	No	Yes
Network activity	No	Yes	Yes	Yes	Yes	Yes
Network statistics	No	No	No	No	Yes	Yes
Observations	1,556	1,556	1,556	1,556	1,556	1,556
R^2	0.185	0.207	0.222	0.242	0.225	0.259
Adjusted R^2	0.179	0.194	0.202	0.217	0.192	0.214
Note:						*p<0.1; **p<0.05; ***p<0.01

Table E.3: Significant variables from models with differing variables include and Lag length 12. Insignificant variables are not displayed. Control variables (network statistics and network activity) are included for differing models.

	Dependent variable:				
	JV				
	(1)	(2)	(3)	(4)	(5)
ANVN11			0.0002***		0.0002**
			(0.0001)		(0.0001)
Constant	−0.001	−0.001	−0.0002	−0.004	−0.007
	(0.001)	(0.001)	(0.001)	(0.010)	(0.010)
Vital nodes	No	Yes	Yes	No	Yes
Non-vital nodes	No	No	Yes	No	Yes
Network activity	Yes	Yes	Yes	Yes	Yes
Network statistics	No	No	No	Yes	Yes
Observations	1,556	1,556	1,556	1,556	1,556
R^2	0.010	0.010	0.020	0.030	0.038
Adjusted R^2	0.002	−0.007	−0.005	−0.004	−0.013
Note:					*p<0.1; **p<0.05; ***p<0.01

and ANVN better explain the dynamics of the integrated variance than the conventional network statistics.

Switching focus towards the jump part of the realised variance, JV, we observe that the models do not explain the jump dynamics well, compare Table E.3. The adjusted R^2 is very low, almost zero, or negative for most models, suggesting that the 5-minute jumps in volatility during the market frenzy are not determined by the WSB dynamics. As we consider the 2 weeks before and after the GME price peak, we have to take into account that many notable events surrounding the GME frenzy saga took place during this time, which caused market disturbances not incited by WSB, such as the ban of Robinhood on buying GME, insults of research firms towards WSB investors or support of GME from public figures such as Elon Musk.

Moving focus towards the 5-minute return dynamics during the market frenzy, we model the dynamics by the past returns, AC, VN, ANVN and network statistics, see Table E.1. We observe that the return dynamics are best explained by their own past dynamics with a small improvement of the adjusted R^2 when adding the network activity to the regression. However, VN, ANVN and the established network statistics do not improve the adjusted R^2 during the market frenzy. We follow from this that the return dynamics during the frenzy are partly explainable by the WSB network activity, however the behavioural change of the VN and ANVN had little effect on the returns.

To sum up, we conclude that during the market frenzy, the integrated variance is explained by the behavioural changes of the VN and ANVN whereas the JV is largely unaffected by the network variables and measures. For the return dynamics, we observe that the

WSB network activity and various network measures improve their explainability, however VN and ANVN do not. This implies that metrics derived from the WSB network do explain the market movements of GMEs stock price, whereas the behavioural changes of the VN and ANVN explain the integrated variance of GME.

F Robustness analysis: Modelling by trading hour

We investigate if the results of our study hold up when we focus the analysis upon the trading hours of NYSE and compute one model per trading hour. By this, we investigate if seasonality of trading is impacting our results. As there are 7 trading hours, this leads to 7 models for each of the time series to be analysed, namely return, integrated volatility and jump volatility. The results of these analysis are displayed in Tables F.1, F.2 and F.3. The results indicate that the change in posting behaviour of the VNs and ANVNs plays an even stronger role when focusing upon the individual trading hours. Across the hours and data series, the R^2 and adj. R^2 are much higher. For the return series, Table F.1, we observe that for all hours but the first and last one, a higher proportion of the variability is explained by the change points of VN and ANVN compared to the network statistics. In all cases, considering the change points and network statistics improves the R^2 and adj. R^2 the most. Notably, the models strongly outperform a standard autoregressive model. For hour 5, the AR model only yields 0.1 adj. R^2 but the model considering AC and the change points of VN and ANVN yields 0.48 adj. R^2.

In terms of modelling the IV and JV by the trading hours, similar results as for the entire time series are observed but with much higher R^2 and adj. R^2, Table F.2. For various hours, the change points of the VN and ANVN add so much explainability of the variability to the base model, that the network statistics only explain very little, consider hours 4, 5, 6, or 7. For hour 4 the change points of the VN and ANVN explain such a large proportion of the variability, $R^2 = 0.9$, whereas the model with the network statistics only explains $R^2 = 0.67$, that it can be concluded the change in behaviour of the users on WSB explain almost the entire variability of the IV. For the JV, Table F.3, we observed that the change points of the VN and ANVN add a lot more explainability to the variability of the time series in the per trading hour analysis compared to the entire time series. For the latter, modelling the JV yielded $R^2 = 0.129$ for the model consisting of change points, network statistics and activity. For the per trading hour analysis, they explain at least 41% of the variability of the system, for hour 4 even 86% of the variability are explained by the change points, network statistics and activity. The model containing only network activity and change points of VN and ANVN explain in most cases, but hour 1, 6, 7, more of the JVs variability than the network statistics.

To sum up, these results show that the change in behaviour from the users on WSB,

measured by the change points of the VN and ANVN, explain a large proportion of the variability of the return series, integrated variance and jump volatility. The results also show that during market trading hours, an even larger proportion of the affected stocks variability can be explained. The latter result is in line with our observation that the hour of the day and trading hours have an effect upon the WSB users posting behaviour, compare section 5.

Table F.1: Return: This table shows that the R^2 and adj. R^2 for the return prediction models regressed onto the observations as by trading hours. Column (1) shows the results for the autoregressive benchmark model, (2)-(6) add variables to the benchmark model. The results show that regardless of the trading hour, the change points of the VN and ANVN strongly improve the modelling of the return series. Notably the R^2 and adj. R^2 are much higher in the hourly regressions than in the models for the entire time series.

		(1)	(2)	(3)	(4)	(5)	(6)
Hour 1	R^2	0.07	0.09	0.1	0.31	0.33	0.49
	adj. R^2	0.03	-0.01	-0.01	0.17	0.1	0.25
Hour 2	R^2	0.1	0.13	0.42	0.58	0.41	0.67
	adj. R^2	0.08	0.08	0.36	0.52	0.29	0.57
Hour 3	R^2	0.29	0.3	0.38	0.46	0.42	0.55
	adj. R^2	0.26	0.25	0.31	0.38	0.31	0.41
Hour 4	R^2	0.52	0.52	0.62	0.65	0.59	0.71
	adj. R^2	0.5	0.49	0.58	0.6	0.51	0.62
Hour 5	R^2	0.13	0.16	0.29	0.55	0.23	0.61
	adj. R^2	0.1	0.1	0.21	0.48	0.07	0.49
Hour 6	R^2	0.16	0.17	0.32	0.36	0.32	0.46
	adj. R^2	0.13	0.11	0.24	0.26	0.17	0.29
Hour 7	R^2	0.18	0.2	0.26	0.29	0.33	0.44
	adj. R^2	0.15	0.14	0.18	0.19	0.19	0.26
	Vital nodes	No	No	Yes	Yes	No	Yes
	Non-vital nodes	No	No	No	Yes	No	Yes
	Network activity	No	Yes	Yes	Yes	Yes	Yes
	Network statistics	No	No	No	No	Yes	Yes

Table F.2: Integrated variance: This table shows the R^2 and adj. R^2 for the integrated variance prediction models regressed onto the observations as by trading hours. Column (1) shows the results for the autoregressive benchmark model, (2)-(6) add variables to the benchmark model. The results show that regardless of the trading hour, the change points of the VN and ANVN strongly improve the modelling of the integrated variance. Notably the R^2 and adj. R^2 are much higher in the hourly regressions than in the models for the entire time series. For Hour 4 we observe a particularly strong improvement from the VN and ANVN change points.

		(1)	(2)	(3)	(4)	(5)	(6)
Hour 1	R^2	0.53	0.56	0.59	0.68	0.65	0.74
	adj. R^2	0.51	0.51	0.54	0.61	0.53	0.61
Hour 2	R^2	0.65	0.67	0.68	0.84	0.74	0.9
	adj. R^2	0.64	0.65	0.65	0.82	0.68	0.87
Hour 3	R^2	0.62	0.63	0.69	0.76	0.72	0.82
	adj. R^2	0.61	0.61	0.66	0.72	0.66	0.77
Hour 4	R^2	0.52	0.53	0.69	0.9	0.67	0.91
	adj. R^2	0.51	0.5	0.65	0.88	0.6	0.89
Hour 5	R^2	0.6	0.61	0.64	0.8	0.73	0.85
	adj. R^2	0.59	0.58	0.6	0.78	0.68	0.8
Hour 6	R^2	0.78	0.8	0.81	0.89	0.85	0.91
	adj. R^2	0.77	0.79	0.79	0.87	0.82	0.88
Hour 7	R^2	0.69	0.71	0.72	0.82	0.75	0.85
	adj. R^2	0.68	0.68	0.69	0.79	0.7	0.8
	Vital nodes	No	No	Yes	Yes	No	Yes
	Non-vital nodes	No	No	No	Yes	No	Yes
	Network activity	No	Yes	Yes	Yes	Yes	Yes
	Network statistics	No	No	No	No	Yes	Yes

Table F.3: Jump volatility: This table shows the R^2 and adj. R^2 for the jump volatility prediction models regressed onto the observations as by trading hours. Column (1) shows the results for the benchmark model with network activity variables, (2)-(5) add variables to the benchmark model. The results show that regardless of the trading hour, the change points of the VN and ANVN strongly improve the modelling of the jump volatility. We observe that contrary to the regression onto the entire time series, the change points of VN and ANVN strongly improve the modelling of the jump volatility and explains parts the network statistic cannot. This observation is particularly pronounced for Hours 2-5, so the morning and midday trading hours. Further, the R^2 and adj. R^2 are much higher in the hourly regressions than in the models for the entire time series.

		(1)	(2)	(3)	(4)	(5)
Hour 1	R^2	0.14	0.15	0.19	0.34	0.41
	adj. R^2	0.1	0.09	0.09	0.17	0.18
Hour 2	R^2	0.14	0.14	0.38	0.37	0.56
	adj. R^2	0.11	0.09	0.31	0.27	0.45
Hour 3	R^2	0.06	0.1	0.31	0.23	0.41
	adj. R^2	0.03	0.04	0.23	0.11	0.26
Hour 4	R^2	0.04	0.49	0.84	0.44	0.86
	adj. R^2	0	0.45	0.82	0.36	0.83
Hour 5	R^2	0.02	0.05	0.42	0.25	0.57
	adj. R^2	-0.02	-0.01	0.36	0.13	0.46
Hour 6	R^2	0.02	0.05	0.11	0.38	0.43
	adj. R^2	-0.02	-0.02	0.01	0.28	0.28
Hour 7	R^2	0.09	0.1	0.23	0.36	0.49
	adj. R^2	0.06	0.04	0.14	0.26	0.36
	Vital nodes	No	Yes	Yes	No	Yes
	Non-vital nodes	No	No	Yes	No	Yes
	Network activity	Yes	Yes	Yes	Yes	Yes
	Network statistics	No	No	No	Yes	Yes